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Automatic Parking Management System using Image-Based License Plate Recognition for Parking Ltd.

In Partial Fulfillment of the Course Requirements for
CAPIT0A

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CHAPTER 1. Project Background

1.1. Introduction

The rapid expansion of urban environments and the continuous rise in vehicle ownership have transformed parking management into a critical logistical challenge that traditional infrastructure struggles to accommodate. Conventional parking facilities commonly rely on manual processes such as physical ticket issuance, guard supervision, and paper-based record-keeping. These approaches are inherently time-consuming, prone to human error, and incapable of providing real-time visibility of parking capacity. As a result, operational inefficiencies emerge in the form of long queues, internal congestion, double parking, and inconsistent transaction handling during peak hours. Moreover, the heavy dependence on manpower for monitoring, billing, and enforcement increases operational costs and limits scalability in high-density urban environments.

To address these limitations, vision-based and computer vision technologies have emerged as a more efficient and scalable alternative to traditional parking systems. As noted by Hadi et al. (2024), computer vision enables real-time monitoring, automated data extraction, and intelligent decision-making without the need for invasive or costly physical sensors. Through camera feeds and image processing techniques, modern systems can accurately detect vehicles, extract license plate information, and analyze parking occupancy. This shift from hardware-dependent systems to

software-driven intelligence reduces maintenance requirements while enabling flexible deployment and enhanced system responsiveness.

Building on these advancements, modern parking infrastructure is transitioning toward a Dynamic Parking Management approach that integrates automation, real-time analytics, and minimal human intervention. Alymani et al. (2025) highlight that combining machine learning with system automation enables fully autonomous access control and operational decision-making. In response to these developments, this study proposes an Automatic Unmanned Parking Management System using Image-Based License Plate Recognition (LPR) integrated with Quick Response (QR) Code Technology, enhanced with real-time spatial allocation, reservation management, and centralized digital monitoring.

Unlike traditional systems, the proposed solution introduces a closed-loop architecture that ensures all vehicle movements are digitally tracked from entry to exit. Upon arrival, vehicles are processed through LPR to capture license plate data, which is then linked to a dynamically generated QR token. This dual-token mechanism establishes a secure and verifiable identity for each vehicle throughout its parking lifecycle. In addition, the system incorporates section-based spatial allocation, where parking slots are assigned intelligently based on real-time occupancy data, reducing internal congestion and eliminating blind entry scenarios.

Furthermore, the integration of QR technology extends beyond authentication by enabling reservation functionality, vendor validation, and flexible payment workflows. Customers may pre-book parking spaces through a web-based application, while commercial tenants can validate transactions digitally without relying on paper-based coupons. The system also supports both automated and attendant-assisted payment processing, ensuring inclusivity for users with limited access to digital payment methods.

Complementing these features is a centralized database and web application, which serves as the system's core management platform. This enables real-time monitoring of parking availability, automated billing based on defined business rules (e.g., grace periods and overnight fees), and generation of audit-ready transaction records. Additionally, vision-based monitoring enforces parking compliance by detecting violations such as unauthorized

parking or incorrect slot usage.

By integrating LPR, QR code technology, real-time occupancy detection, and web-based management, the proposed system delivers a secure, efficient, and scalable parking ecosystem. It not only addresses the operational inefficiencies and revenue leakage present in traditional systems but also enhances user experience through guided parking, faster transactions, and transparent monitoring.

1.2. Industry/Sector/Organizational Profile

1.2.1. The Parking Industry Landscape

The parking industry in the Philippines is undergoing a critical transition driven by rapid urbanization and a surge in vehicle ownership. According to the Chamber of Automotive Manufacturers of the Philippines, Inc. (CAMPI), vehicle sales hit a record high of 467,252 units in 2024, representing an 8.7% increase from the previous year (MarkLines, 2025). This influx of private vehicles has outpaced the development of road and parking infrastructure, creating a severe deficit in available parking spaces in major urban centers like Cabanatuan City.

This imbalance leads to "cruising," where drivers circle indefinitely to find a spot, contributing to traffic congestion and carbon emissions. While the market is shifting toward Smart Parking Systems, a vast majority of local facilities—particularly small-to-medium enterprises (SMEs)—continue to rely on "legacy" manual systems. These manual operations are increasingly viewed as the bottleneck in urban mobility (Technavio, 2026).

1.2.2. Organizational Profile: Parking Ltd

The subject of this study is Parking Ltd, a privately owned commercial parking facility

located in the high-density district of Cabanatuan City. The facility is strategically situated directly in front of Doctor's Hospital and is surrounded by major educational institutions, specifically the College of the Immaculate Conception (CIC) and Our Lady of Fatima University (OLFU) Cabanatuan Campus (Loreto, personal communication, February 15, 2026).

Strategic Location Analysis: The facility's proximity to a major hospital and two universities creates a unique "mixed-use" traffic profile:

High Turnover (Daytime): Students, faculty, and hospital visitors drive the peak congestion hours between 10:00 AM and 3:00 PM.

Long-Term/Overnight: Hospital watchers and staff contribute to the demand for overnight parking. This location explains the critical issue of "double parking" and capacity overload mentioned in the interviews, as the demand from these three institutions frequently exceeds the lot's supply.

Operational Capacities: Based on interview data, Parking Ltd has a total capacity of approximately 100 vehicles.

Fee Structure: A flat rate of ₱30.00 is charged for day parking, while an "Overnight Fee" of ₱100.00 applies to vehicles remaining past 11:00 PM.

1.2.3. Organizational Structure

Parking Ltd operates under a sole proprietorship structure, ensuring centralized decision-making but limited manpower redundancy. The facility's ecosystem includes the administration, operational staff, and commercial tenants.

Proprietor / Owner (Roro James Cruz): Mr. Cruz oversees the general business strategy and revenue auditing. As the sole owner, he is the primary decision-maker regarding infrastructure upgrades, such as the proposed transition to an automated system.

Parking Attendants / Maintenance (Jomar & Loreto): Daily operations are managed by two rotating attendants. Their duties include manual barrier operation, cash collection, and physical logbook recording.

Dependence Risk: The interviews revealed that attendants are also responsible for custodial tasks (e.g., restroom cleaning). When an attendant is occupied with maintenance or personal breaks, the "Open Barrier" system becomes vulnerable to unauthorized exits and revenue leakage.

Commercial Tenants / Vendors: The facility leases commercial space to several independent businesses operating within the parking perimeter, specifically Joey's Restaurant, 228 Cafe, Chick Chicken, and Detalye Car Detailing Services. Because these vendors are direct tenants of Mr. Cruz, they operate under his administrative policies. This hierarchical relationship ensures that if the owner issues a memorandum requiring the implementation of new system hardware—such as thermal printers for generating "drive-thru" QR validation codes—each vendor is obligated to procure and operate this equipment independently to support the parking lot's automated bypass feature.

1.3. Description of Existing System/Infrastructure

Located in Cabanatuan City, the current parking facility is an open-air, single-level lot divided into two distinct terrains: a paved parking section and an unpaved rough road area. Without a roof or structural flooring, the facility operates as a manual, attendant-managed system with a distinct "open-loop" architecture. It relies entirely on human vigilance for access control, fee calculation, and record-keeping, utilizing basic electromechanical barriers without any integration of digital sensors or automated logic.

1.3.1. Operational Workflow

1.3.1.1 Entry Phase (Uncontrolled Access)

Upon arrival, vehicles encounter an entrance boom barrier that is kept in a permanent "Open" (up) position during operating hours. No transaction occurs at the point of entry—drivers are not issued physical tickets, RFID cards, or any proof of arrival time. Consequently, the facility has no mechanism to validate the exact time a vehicle entered the premises.

1.3.1.2 Parking Phase (Blind Capacity Management)

Drivers navigate the parking area independently. There are no sensors or counting mechanisms to track occupancy. Capacity management is reactive; attendants only realize the lot is at its maximum capacity (approximately 100 vehicles) when visual congestion occurs or when drivers begin double-parking in unauthorized zones.

1.3.1.3 Exit and Payment Phase (Manual Processing)

The exit process acts as the single point of control. Vehicles must stop at the guardhouse, where the exit barrier is kept "Closed" (down).

1. **Identification:** The attendant visually inspects the vehicle.

2. **Fee Calculation:** A flat rate of ₱30.00 is charged for standard parking. If the vehicle remains past 11:00 PM, it is manually classified as "Overnight" and charged ₱100.00. However, a grace period of 1 hour or until 12:00 AM is given to customers.

3. **Recording:** After payment is received, the attendant manually writes the license plate number into a physical logbook.

4. **Actuation:** The attendant presses a manual push-button to lift the barrier and release the vehicle.

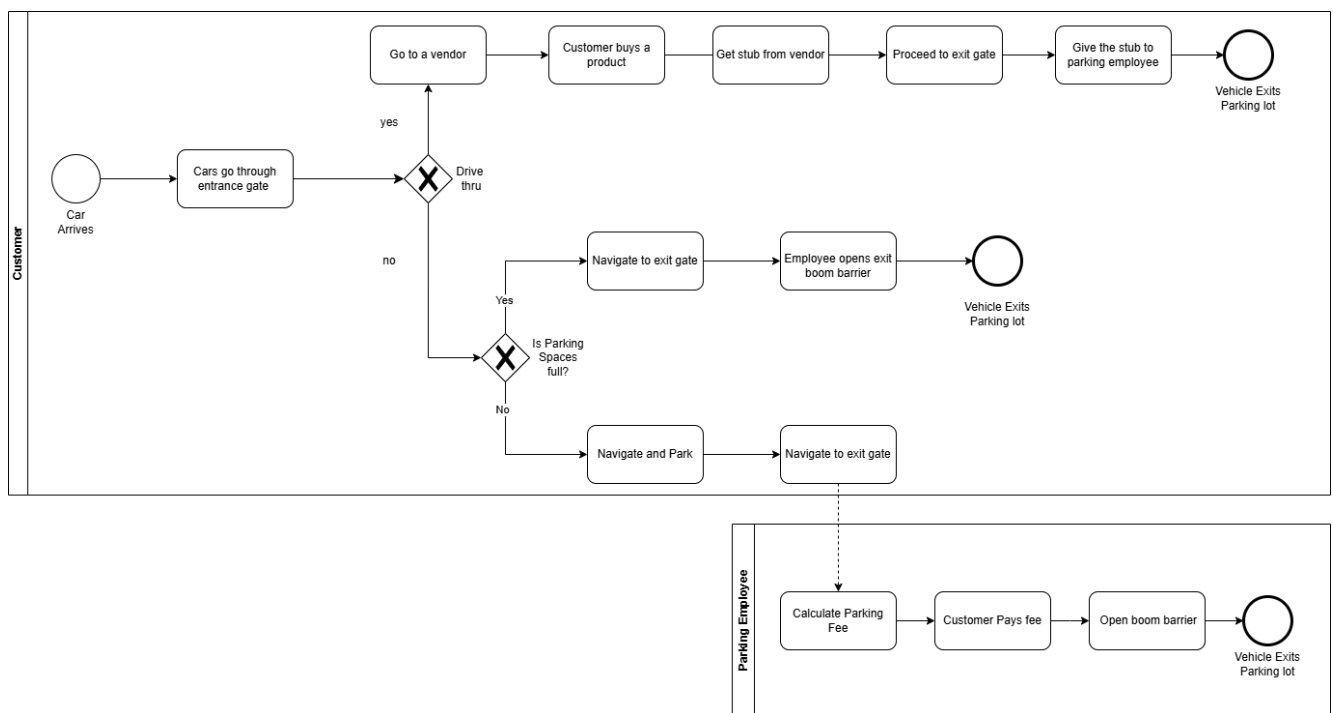


Figure 1.0: BPMN of the Current Workflow of Parking LTD.

1.3.2. Hardware and Infrastructure

The facility's infrastructure is comprised of standalone, non-networked components:

Barriers: Two electromechanical boom barriers (one for Entry, one for Exit) operated via manual push-button consoles inside the guardhouse.

Power Supply: Standard electrical connectivity is available at the gate and

guardhouse.

Network: There is no internet connectivity or local area network (LAN) at the gate; internet access is limited to the main administrative office, isolating the gate operations from real-time administrative oversight.

1.3.3. Data Management and Security

All parking data is stored in handwritten notebooks. Standard entries are logged in one notebook, while overnight vehicles are recorded in a separate log. This makes data retrieval (searching for past records) slow and prone to human error.

Since the entrance barrier is left open to facilitate flow, the system suffers from a critical security flaw. "Unauthorized Exits" occur when vehicles bypass the exit guardhouse by driving out through the open entrance lane (counter-flowing). Furthermore, without an entry log, there is no way to verify how long a "pass-through" vehicle has actually been inside the facility, leading to potential revenue leakage.

1.4. Statement of the Problems and Opportunities

1.4.1. Specific Problem #1 – Operational Inefficiency and Lack of Real-Time Capacity Management

The current parking facility suffers from a lack of data visibility regarding occupancy. Drivers enter the facility blindly, unaware if parking spots are available, leading to "cruising" (circling) and double-parking that obstructs flow. Furthermore, the exit process is heavily bottlenecked by manual recording, where attendants must physically write license plate numbers into a logbook before releasing the vehicle.

1.4.1.1. 5 Whys Analysis

Why is there congestion and "double parking" inside the facility?

Drivers continue to enter the facility even when it is at full capacity (100 vehicles) because there is no mechanism to stop them or warn them at the gate. This forces drivers to circle continuously or park in unauthorized areas, creating internal gridlock.

Why is there no warning or entry restriction when the lot is full?

The facility lacks a real-time counting system. The current method relies on manual observation, which cannot instantaneously calculate the net difference between entries and exits to determine available slots.

Why is the exit process slow, causing queues during peak hours (10 AM - 3 PM)?

The transaction cycle is manually dependent. The attendant must visually inspect the plate, calculate the duration based on memory or a disparate "overnight" logbook, and physically write the details into a notebook. This manual serialization takes significantly longer than the 3-second processing time achievable by automated systems (Rani et al., 2024).

Why is the system reliant on manual logbooks instead of digital logs?

There is no integrated software to capture and store vehicle data automatically.

The infrastructure acts as "siloe hardware," where the barrier gate and the recording process are completely verified by human effort rather than digital triggers (Pradhan et al., 2025).

Why has the facility not adopted automated counting or logging?

The root cause is the reliance on legacy operational models due to the perceived complexity of computer vision. Management relies on human attendants (e.g., Jomar and Loreto) to manage complex variables like "overnight parking" because they lack a "Dynamic Parking Management" framework that can handle these rules algorithmically (Alymani et al., 2025).

1.4.1.2. Root Cause Evaluation

The root cause is the lack of digital synchronization between physical flow and data flow. Because the barrier operation is not tied to a database, the system cannot count cars (to prevent overcrowding) nor can it process exits quickly (to prevent queues). The reliance on a "notebook" database makes real-time capacity management impossible.

1.4.1.3. Opportunity

This presents an opportunity to deploy an automated Occupancy Monitoring System. By utilizing YOLOv8 for vehicle detection (Satya et al., 2025), the system can maintain a real-time count of vehicles (Entries - Exits). This allows for the installation of a "FULL" indicator at the

entrance to prevent overcrowding and replaces the "notebook" log with a searchable SQL database for instant exit processing (Ezginci & Karadaş, 2019).

1.4.2. Specific Problem #2 – Risk of Revenue Leakage via "Open Loop" Security

The most critical financial risk identified is the "Open Loop" nature of the current infrastructure. As initially confirmed by the interview, entrance barriers are often left open during operating hours to maintain traffic flow. To empirically validate this vulnerability, an investigation of the facility's CCTV footage was conducted covering one standard weekday and one weekend. The observational data explicitly captured a total of three unpaid departures—specifically, two unauthorized exits during the weekday and one unauthorized exit during the weekend (see *Appendices F, G, and I*). This physical evidence proves that drivers can easily exploit the open entrance lane or leave without payment if the attendant is momentarily distracted, leading to unrecorded transactions and direct revenue loss.

1.4.2.1. 5 Whys Analysis

Why are there unauthorized exits and uncollected parking fees?

Vehicles are able to leave the premises without a validated transaction. Drivers can exploit the open entrance barrier or pass through the exit when the attendant is occupied, as there is no physical interlocking mechanism to prevent this.

Why are the entrance barriers left open or easily bypassed?

The current barriers operate on a manual switch or are left in an "up" position to

facilitate faster entry. There is no system to automatically trigger the barrier only when a license plate is successfully captured and logged.

Why is there no strict "No Plate, No Entry" policy enforced by the hardware?

The facility lacks an Automatic License Plate Recognition (ALPR) system to act as the "gatekeeper." Without a camera to verify identity, the barrier cannot distinguish between a legitimate customer and a pass-through vehicle, forcing attendants to leave gates open to avoid blocking traffic.

Why is it difficult to track "pass-through" vehicles or "overnight" stayers?

The current system relies on disparate recording methods (e.g., a separate notebook for overnight parking). There is no "single source of truth" that links an Entry Timestamp to an Exit Timestamp. If a car enters at 10 AM and claims to have entered at 2 PM, the attendant has no quick way to verify this without searching through pages of handwritten notes.

Why is the system vulnerable to these manual loopholes?

The root cause is the lack of an automated authentication handshake. In secure systems, the license plate serves as the "password" to open the gate (Al-batat et al., 2022). Without this digital key, physical security is entirely dependent on the vigilance of the staff, which is prone to error and fatigue.

1.4.2.2. Root Cause Evaluation

The evaluation reveals that Revenue Leakage is a symptom of Uncontrolled Access. The failure to enforce a "Closed Gate" policy (where the gate is always down until a plate is scanned) creates a physical loophole that results in financial loss.

1.4.2.3. Opportunity

The study identifies an opportunity to implement a "Closed Loop" Unmanned System. By integrating YOLOv8 for detection and SQLite for data storage (Pawlaszczyk, 2022), the system can enforce a strict protocol: The barrier only opens if the plate is logged. Furthermore, the system can automatically apply "Grace Period" logic (e.g., free exit if duration < 10 mins) and "Overnight" calculation, removing the need for manual decision-making and closing the revenue gap (Zakaria et al., 2025).

1.4.3 Specific Problem #3 - Lack of Record Management

The current facility suffers from a severe data deficit regarding vehicle tracking and historical auditing. Based on the interview, the administration possesses absolutely no record of customers entering or utilizing the facility, except for a physical notebook where attendants manually write down the license plates of vehicles only when they pay at the exit. Consequently, if a vehicle enters the premises but bypasses the exit, leaves during a guard's shift change, or is involved in a security incident within the lot, there is zero documented proof that the vehicle was ever there. Research by Parmar et al. (2020) highlights that legacy parking systems relying on

physical paper logs often suffer from severe data loss and lack of traceability, leaving facility owners completely blind to unbilled vehicles and incapable of performing accurate post-incident audits.

1.4.3.1 5 Whys Analysis

Why does the facility lack reliable vehicle records?

Because the administration does not maintain a complete and consistent log of all vehicles that enter and use the facility. As a result, only a small portion of actual parking activity is formally recorded.

Why is the vehicle log incomplete and inconsistent?

Because the current process depends on manual encoding in a physical notebook, and only after payment at the exit. This approach excludes vehicles that do not complete the standard payment process or are missed during busy periods or shift transitions.

Why are records only created during payment at the exit?

Because the current recording practice is tied to the collection process rather than to vehicle entry, occupancy, and exit monitoring. Therefore, vehicles that leave without normal exit processing or are missed during shift changes may never be documented.

Why is the recording practice tied only to payment collection?

Because the facility follows a manual, transaction based workflow that prioritizes

fee collection over continuous historical documentation. This creates gaps in traceability, especially for vehicles that bypass the normal exit procedure or are involved in incidents inside the premises.

Why does the facility rely on a transaction-based workflow instead of continuous documentation?

Because there is no formal record management policy or structured operational process that requires the capture of vehicle data from entry to exit. Consequently, the facility's records remain fragmented, incomplete, and unsuitable for accurate auditing or incident verification.

1.4.3.2 Root Cause Evaluation

The root cause of the record management problem is the facility's reliance on a manual, transaction based documentation process that captures vehicle information only at the point of payment. Because record taking is not integrated into the full parking workflow, the system does not preserve vehicle presence from entry to exit. This means that only vehicles that complete the standard exit transaction are recorded, while those that bypass payment, leave during shift changes, or are involved in incidents may remain undocumented.

1.4.3.3 Opportunity

The identified problem presents an opportunity to improve the facility's operational accountability through the implementation of a more systematic parking record management approach. By establishing a process that captures vehicle related data throughout the parking lifecycle, the facility can move from incomplete manual documentation to a more reliable and

traceable recording system. This would allow the administration to maintain historical records of vehicle presence, support post-incident verification, and improve oversight of parking activity.

1.5. Objectives of the Study

1.5.1. General Objective

To design and develop a parking management system for Parking Ltd. that addresses inefficiencies in vehicle monitoring, ticket validation, billing accuracy, revenue control, and exit verification in the current parking operations.

1.5.2. Specific Objectives

- To reduce entry and exit processing delays by replacing manual ticket handling with a more efficient digital identification system.
- To enhance parking security by implementing a reliable vehicle verification mechanism that ensures proper vehicle-to-ticket matching.
- To ensure accurate computation of parking fees based on established business rules, including grace periods and overnight charges.
- To minimize revenue loss and prevent unauthorized exits through real-time payment validation prior

to vehicle release.

- To improve vendor validation procedures by enabling accredited establishments to authorize legitimate parking fee exemptions.
- To strengthen transaction monitoring and reporting through centralized digital record management.

1.6 Design Architecture

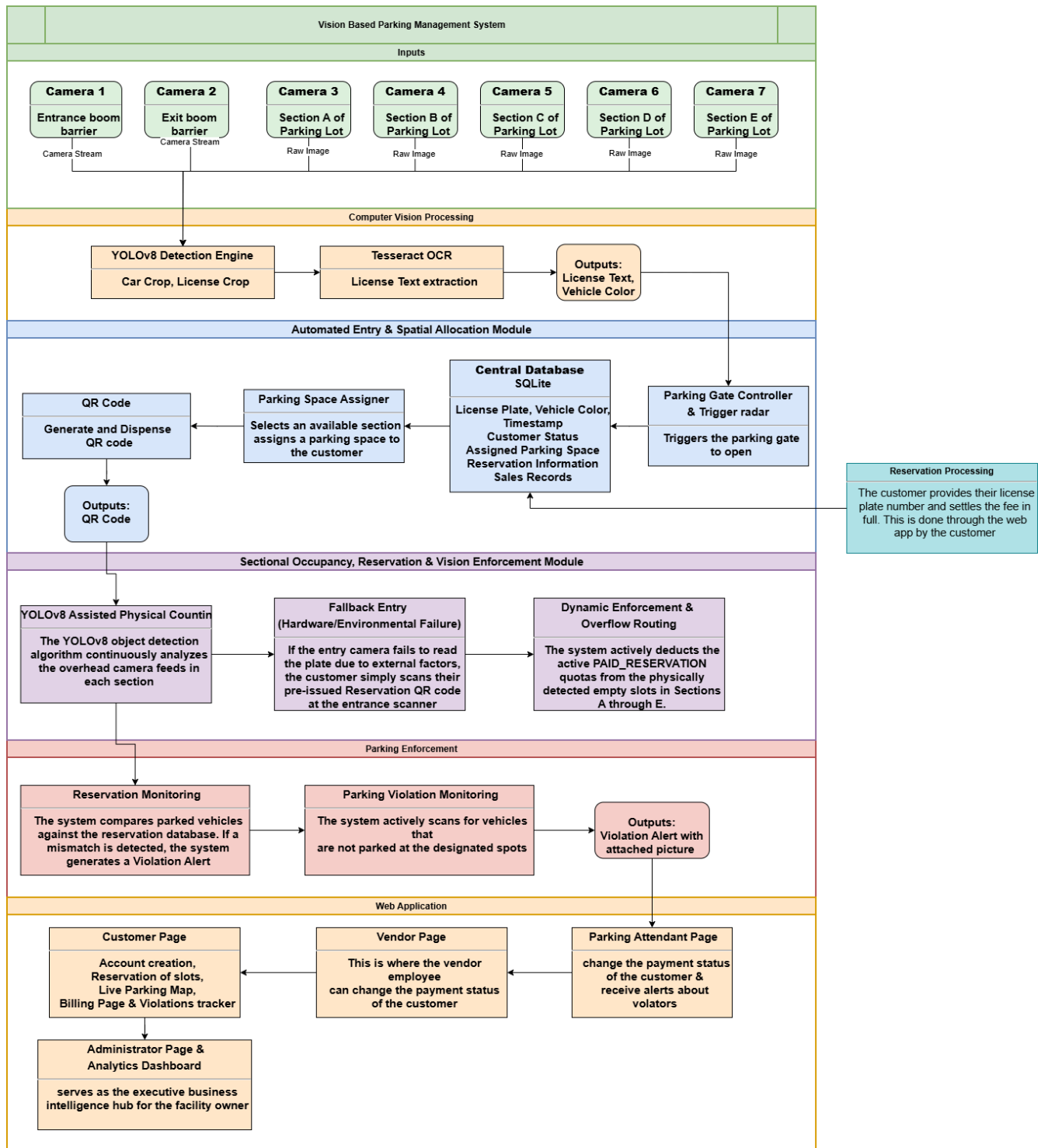


Figure 1.1: Design Architecture of the Automatic Unmanned Parking Management System

Inputs

This tier serves as the foundational interaction point between the physical parking environment and your digital system. It primarily consists of the hardware components deployed at the entry and exit points of the parking facility. High-resolution CCTV cameras are strategically positioned to capture clear, unobstructed images of approaching vehicles and their license plates. Cameras will also be positioned in key locations overlooking the different sections of the parking lot. These cameras will be integrated to a local PC that immediately ingest the raw video feed. By covering both the primary access points and the internal parking areas, this tier ensures there are no blind spots in the system's operational view. The continuous streams and raw images collected here serve as the critical first input for all subsequent automated processes.

Computer Vision Processing

Functioning as the primary analytical engine, this tier transforms the raw visual feeds from the inputs layer into structured, actionable data. It leverages the highly efficient YOLOv8 object detection engine to instantly analyze incoming frames, isolating the vehicle and cropping the exact location of the license plate. Once the plate is cropped, the system feeds this isolated image into Tesseract OCR, a powerful optical character recognition tool. Tesseract then reads the visual data and extracts the alphanumeric characters of the license plate into standard text format. Additionally, this processing layer analyzes the broader car crop to determine the physical color of the vehicle. The final outputs of this tier, the extracted license text and the vehicle color are then neatly packaged and forwarded to the central database.

Automated Entry & Spatial Allocation Module

This middle tier functions as the operational brain of the system, governing the initial vehicle onboarding and physical access protocols. At its core lies the SQLite Central Database, which securely stores and updates critical information such as the license plate, vehicle color, timestamp, assigned spaces, and sales records. When a vehicle is processed at the entrance, the system triggers the generation of a unique QR code tied directly to that specific license plate.

As a vehicle approaches the entry point, the radar sensor detects its presence. This triggers the LPR camera to capture the vehicle's license plate and open the barrier. Simultaneously, the Parking Space Assigner queries the database to select an available section and officially assigns a parking space to the incoming customer. As the vehicle passes through, another radar sensor continuously monitors the area directly under the barrier arm. Once the radar confirms the vehicle has completely cleared the zone, it sends an automatic signal to close the barrier, securing the facility for the next user.

Rule Based Algorithm

- **Rule 1: Reservation Protocol**
 - *Condition:* IF the LPR matches a plate with an active reservation.
 - *Action:* THEN bypass general allocation and assign the exact pre-booked slot. The entrance display directs the driver to that specific slot.
- **Rule 2: Special Category Protocol (PWD/VIP)**
 - *Condition:* IF the LPR matches a registered user flagged as PWD/VIP (and has no specific reservation).
 - *Action:* THEN scan **Zone A**. Assign the closest available slot to the entrance. IF Zone A is full, proceed to Rule 3.
- **Rule 3: Standard Sequential Allocation (Walk-ins)**
 - *Condition:* IF the vehicle is a standard walk-in.
 - *Action:* THEN the system scans for the nearest available slot to the entrance to prevent queue buildup.
 - Check **Zone B** first. Assign the first available slot.
 - IF Zone B is full, check **Zone C**.
 - IF Zone C is full, check **Zone D**.
- **Rule 4: Capacity Management**
 - *Condition:* IF Zones A, B, C, and D, E are all at 100% capacity.
 - *Action:* THEN the entrance barrier will still open, the web app displays "Proceed to Area E (Rough Road section)."

Sectional Occupancy, Reservation & Vision Enforcement Module

Once a vehicle is inside the facility, this tier takes over to actively manage physical space utilization and handle operational edge cases. It leverages YOLOv8-assisted physical counting to continuously analyze the overhead camera feeds from Sections A through E, maintaining an accurate, real-time tally of occupied spaces. To account for hardware or environmental failures at the entry cameras (such as heavy rain or glare), a robust Fallback Entry mechanism allows users to manually scan a pre-issued Reservation QR code at the gate scanner. Furthermore, the Dynamic Enforcement and Overflow Routing protocol

actively reconciles the digital ledger with any changes in capacity. It continuously deducts the active, paid reservation quotas from the physically detected empty slots across the lot. This ensures that the system never double-books a space that has been guaranteed to a reserved customer, even if that customer has not yet physically arrived.

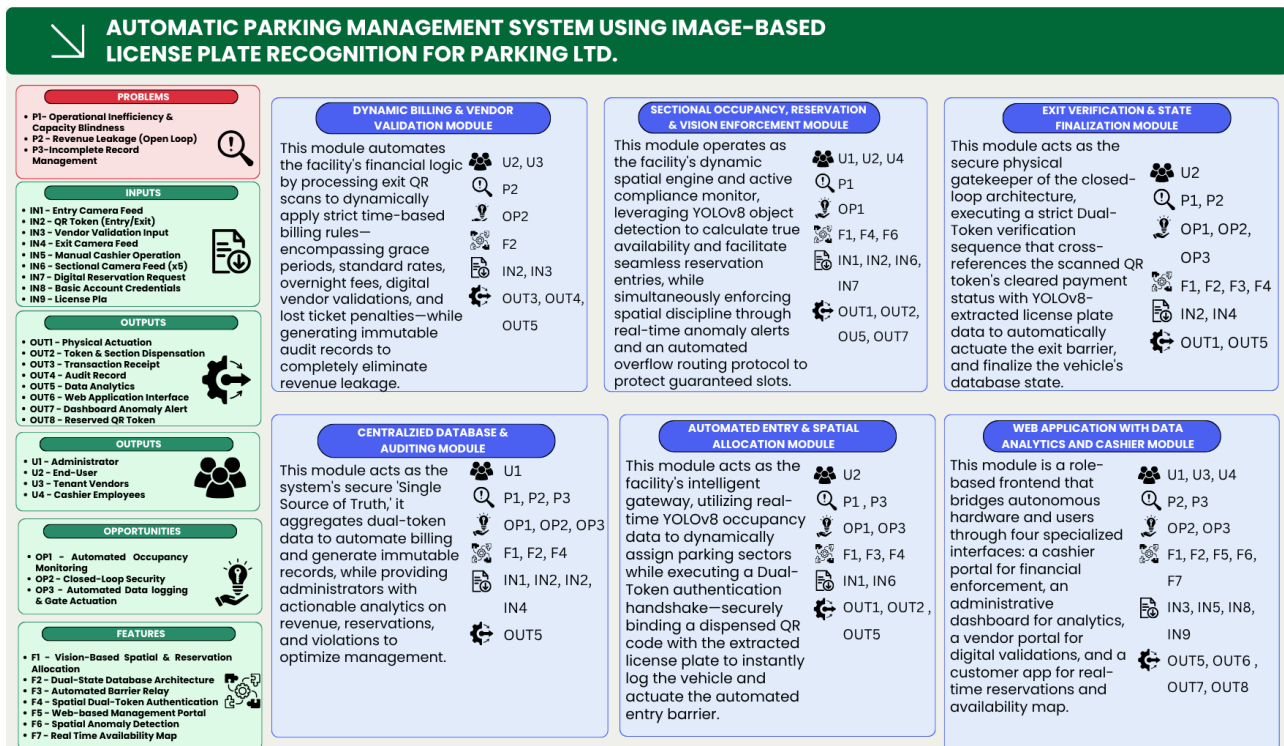
Parking Enforcement

This automated security tier acts as the facility's digital patrol, ensuring strict compliance with the allocated parking assignments. It utilizes the Reservation Monitoring feature to continuously cross-reference physically parked vehicles against the active records in the SQLite database. If the system detects that a vehicle is parked in a spot reserved for someone else, it immediately registers a mismatch. Simultaneously, the Parking Violation Monitoring system actively scans the entire lot for any vehicles parked completely outside of designated areas or in restricted zones. When either of these infractions occurs, the system automatically generates a formal Violation Alert. This alert is bundled with an attached picture captured by the overhead cameras to provide undeniable visual proof of the infraction.

Web Application

The final tier serves as the primary human-computer interface, translating complex backend operations into accessible tools for various stakeholders. It features an Administrator Page and Analytics Dashboard, which functions as the executive business intelligence hub, giving facility owners comprehensive oversight over revenue, utilization metrics, and system health. For third-party integration, the Vendor Page offers a dedicated portal where tenant employees can easily locate a customer's record and update their payment status, securely closing the revenue loop. The Parking Attendant Page equips on-site staff with the necessary controls to manually alter payment statuses and immediately receive the violation alerts generated by the enforcement tier. Finally, the customer page allows people to create accounts, view slot availability, reserve slots, and resolve any violation committed. By compartmentalizing these tools into role-specific web portals, the system guarantees secure and relevant access for every type of user.

1.7. Conceptual Framework



1.7.1. Conceptual Framework Discussion

The conceptual framework of this study relies on a **Dual-State Logic Model** integrated with an **Automated Hardware Relay**. Unlike the existing manual system, where operation depends on human physical labor (writing in notebooks and toggling switches), the proposed system shifts to a fully digital workflow.

The framework tracks two distinct and concurrent attributes for every vehicle to address the identified operational gaps:

1. **Physical Presence (Parked vs. Exited):** This variable governs the **Capacity Counter**, solving the "blind entry" and congestion issues (**P1**).
2. **Financial Status (Unpaid, Paid, or Bypassed):** This variable governs the **Exit Barrier**, acting as the authorization key to prevent revenue leakage (**P2**).

Furthermore, the integration of **IoT Relay Triggers** eliminates the need for manual barrier actuation, directly addressing the operational fatigue experienced by attendants (**P3**). This architectural separation ensures that a vehicle can be legally "Paid" while still being physically "Parked" (e.g., during the grace period), preventing capacity drift while securing revenue.

1.7.2. System Variables

1.7.2.1 Problems

P1 - Operational Inefficiency & Capacity Blindness: The lack of real-time occupancy data causes drivers to enter a full facility, creating internal gridlock and "double parking."

P2 - Revenue Leakage (Open Loop): The "Open Loop" nature of the current system allows unauthorized exits and unverified "pass-through" claims due to open barriers.

P3 - Incomplete Record Management: The absence of automated entry auditing creates a "data black hole," leaving the facility completely reliant on easily misplaced manual notebooks that only record vehicles at the exit.

1.7.2.2 Opportunities

OP1 - Automated Occupancy Monitoring: Utilizing entry/exit sensors to maintain a real-time count (Entries - Exits) to prevent overcrowding and inform drivers.

OP2 - Closed-Loop Security: Enforcing a "Closed Gate" policy where the barrier only opens upon digital verification of payment.

OP3 - Automated Data Logging & Gate Actuation: Implementing hardware relays and database triggers to instantly capture entry records and actuate barriers, eliminating manual traffic bottlenecks and ensuring 100% entry traceability.

1.7.2.3 Features

F1 - Vision-Based Spatial & Reservation Allocation: Occupancy is dynamically managed by cross-referencing YOLOv8 object detection across five sectional camera feeds against active digital reservation quotas to compute true slot availability.

F2 - Dual-State Database Architecture: Implementation of separate database columns for Location_Status and Payment_Status.

F3 - Automated Barrier Relay: The entrance radar sensor will automatically raise the parking barrier once it detects a vehicle.

F4 - Spatial Dual-Token Authentication: The systematic pairing of a physical/digital Quick Response (QR) code—which now includes a designated Parking Section assignment—with the visual License Plate Recognition (LPR) data.

F5 - Web-based Management Portal: A centralized browser-accessible interface that facilitates both real-time operational control for staff, long-term strategic analysis for the administrator, and a front-end interface for customers.

F6 - Spatial Anomaly Detection: The capability of the system to cross-reference mapped facility boundaries against real-time YOLOv8 bounding boxes to detect vehicles parking in unauthorized zones or violating their assigned spatial allocation.

F7 - Real Time Availability Map - The web app will display a digital rendering of the parking lot showing which slots are occupied and which ones are free.

1.7.2.4 Users

U1- Administrator: Parking Ltd Owner (Roro James Cruz).

U2- End-User: Transient Drivers and Overnight Parkers.

U3- Tenant Vendors: Commercial establishments (e.g., Joey's Restaurant) issuing validation.

U4- Cashier Employees: Employees assigned to the payment counter.

1.7.2.5 Inputs

IN1 - Entry Camera Feed: Visual LPR data capturing the license plate upon arrival, securely linked to the exact entry timestamp.

IN2 - QR Token (Entry/Exit): The unique QR code dispensed at entry (associated with the entry timestamp and assigned section). It is scanned by the customer at the exit terminal to calculate total duration, or scanned at the entrance terminal as a fallback access key for pre-paid reservations.

IN3 - Vendor Validation Input: The system triggers when a tenant vendor scans the ENTRY QR and securely updates its database status to "PAID_VENDOR" for the exit scanner.

IN4 - Exit Camera Feed: Visual LPR data capturing the plate at the exit gate for cross-referencing, securely linked to the exact exit timestamp to validate the time spent.

IN5 - Manual Cashier Operation: An input stream from the web portal where an attendant manually verifies cash receipt and triggers a status update for a specific QR/Plate record.

IN6 - Sectional Camera Feeds (x5): Continuous visual data streams monitoring Sections A through E of the parking facility. These feeds utilize YOLOv8 to detect the presence of parked vehicles and compute available slots per zone.

IN7 - Digital Reservation Request: An input stream originating from the Web-Based Portal (F5) where users pre-book a parking schedule, generating an advanced QR token.

IN8 - Basic Account Credentials: A customer's email and password to be used in creating an account for parking reservation.

IN9 - License Plate Number: A customer can enter their license number into the web app to see if they have any outstanding bills or violations.

1.7.2.6 Outputs

OUT1 - Physical Actuation: Barrier Gate Open/Close signals via Relay.

OUT2 - Token & Section Dispensation: The generation and printing/display of the Entry QR Code, which explicitly includes the system-assigned parking area (e.g., "Assigned: Section A").

OUT3 - Transaction Receipt: Digital or printed proof of payment at the exit scanner.

OUT4 - Audit Record: Structured database row for revenue reporting.

OUT5 - Data Analytics: A visualization of relevant data into charts and tables for the administrative view.

OUT6 - Web Application Interface: The rendered frontend portals (Cashier, Admin, Vendor, Customer) that facilitate human-system interaction.

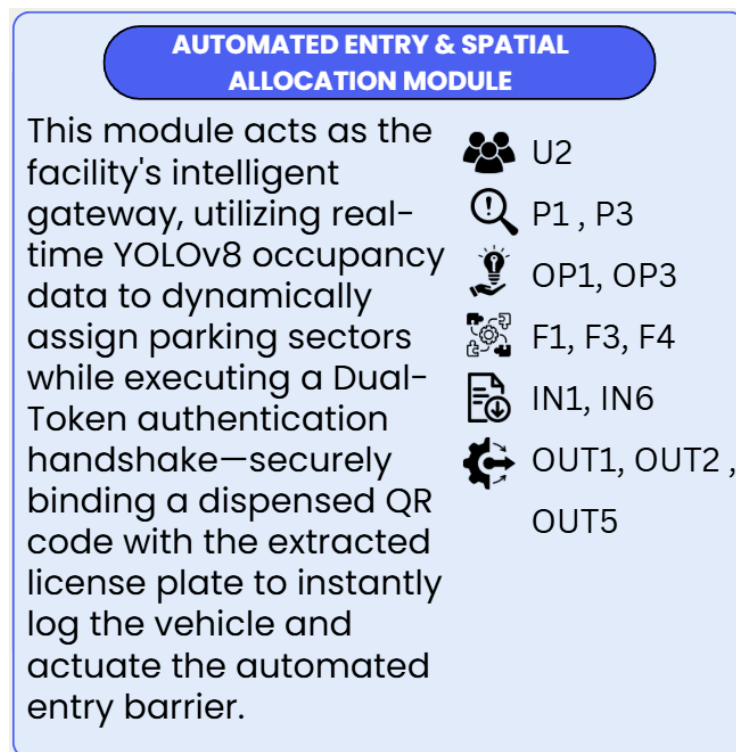
OUT7 - Dashboard Anomaly Alert: A real-time visual and auditory notification pushed to the Web-Based Management Portal to alert attendants of a spatial violation, requiring human intervention.

OUT8 - Reserved QR Token: A digital QR Code given once a customer reserved a spot via the Web App.

1.7.3. Core Modules

The system is engineered into four integrated modules that manage the vehicle's state transitions, integrating both QR and LPR technologies to address the specific problems identified.

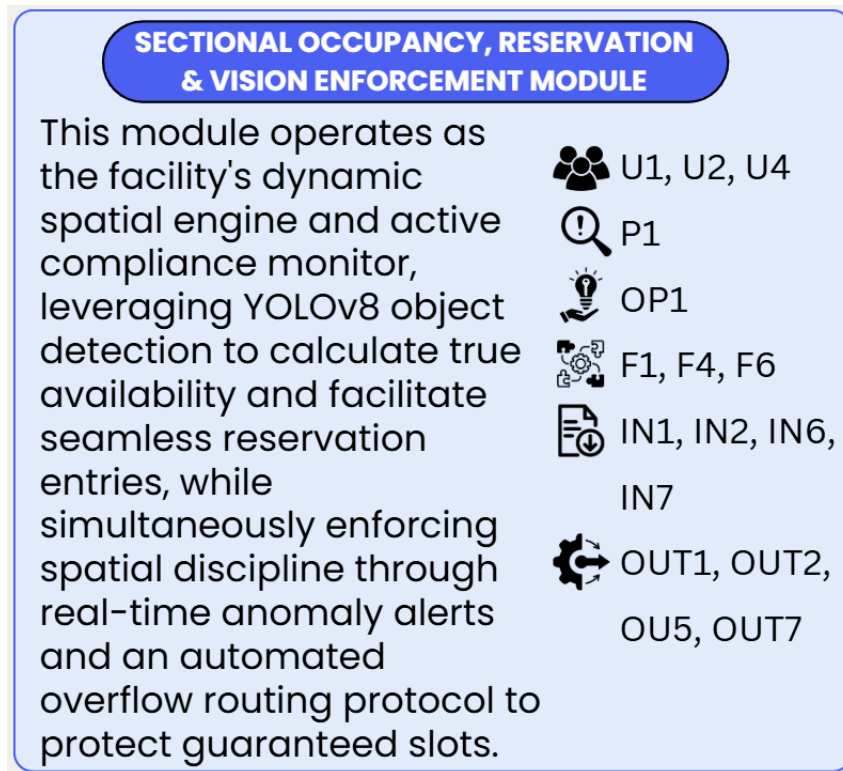
1.7.3.1 Automated Entry & Spatial Allocation Module



Description:

This module serves as the facility's primary gateway, initializing the vehicle's digital lifecycle and actively mitigating internal congestion through intelligent spatial allocation. Upon a vehicle's arrival, the Python-based backend queries the Sectional Occupancy Module's YOLOv8 camera data to dynamically assign the driver to the most vacant paved sector (Sections A-E), or defaults to the unmonitored rough road (Section F) if the main facility is at maximum capacity. As a vehicle approaches the entry point, the radar sensor detects its presence and the system performs a Dual-Token authentication handshake: it dispenses a Quick Response (QR) token explicitly printed with the assigned parking section, while simultaneously utilizing the entry CCTV camera, OpenCV preprocessing, and Tesseract OCR to extract the vehicle's alphanumeric license plate. Finally, these two tokens—the QR ID and the visual plate string—are securely bound together in a centralized SQLite database alongside the exact entry timestamp and assigned section, instantly triggering an edge microcontroller relay to actuate the entrance boom barrier and allowing the driver to proceed directly to their designated zone while permanently resolving the facility's entry data deficit.

1.7.3.2 Sectional Occupancy, Reservation & Vision Enforcement Module



Description:

This module operates as the facility's dynamic spatial engine and active compliance monitor, continuously synchronizing real-time physical camera data with strict digital reservation business rules to rigidly enforce space allocation across six distinct parking zones. The reservation lifecycle is initiated strictly through the client-side web application, where a customer selects an available physical slot and inputs their specific alphanumeric license plate for a duration of one to three days. To optimize slot turnover and prevent monopolization, the system enforces rigid booking parameters: reservations are restricted exclusively to the current week, single user accounts are systematically limited to reserving one slot for one designated vehicle, and all transactions are strictly non-refundable upon payment. Once the digital fee is cleared, the database inextricably binds a 'RESERVED' status to both the specific license plate and the chosen slot, instantly generating a virtual QR token for the user.

By monitoring the paved Sections A through E using YOLOv8 object detection, the system calculates "True Availability"—mathematically deducting both physically parked vehicles and these active prepaid reservations from the maximum sectoral capacity. When a reserved customer arrives, the system prioritizes a frictionless License Plate Recognition (LPR) match via the entry camera to instantly actuate the barrier, while providing the virtual QR token as a robust fallback to bypass redundant physical token dispensation during environmental camera failures.

Beyond capacity calculations, the module's vision architecture actively enforces spatial discipline and reservation integrity. It continuously cross-references physical vehicle bounding boxes and extracted plates against their database-assigned sections. If the system detects a license plate mismatch at a specific reserved slot, or identifies a vehicle persisting within virtual "No Parking" zones (e.g., driveways, fire lanes), it instantly pushes a plate-specific Dashboard Anomaly Alert to the web app. This real-time notification empowers attendants to proactively address the exact violation. Concurrently, to protect these guaranteed reservation slots during peak congestion, the module automatically triggers an Overflow Protection Protocol when the true availability of monitored sections drops to zero. This protocol routes all subsequent walk-in vehicles directly to the unmonitored Section F rough road area via redirection prompts printed on their dispensed QR tokens, securely logging all state transitions into the central database.

1.7.3.3 Dynamic Billing & Vendor Validation Module

**DYNAMIC BILLING & VENDOR
VALIDATION MODULE**

This module automates the facility's financial logic by processing exit QR scans to dynamically apply strict time-based billing rules—encompassing grace periods, standard rates, overnight fees, digital vendor validations, and lost ticket penalties—while generating immutable audit records to completely eliminate revenue leakage.

 U2, U3

 P2

 OP2

 F2

 IN2, IN3

 OUT3, OUT4,
OUT5

Description:

This module governs the time-based financial logic and third-party validations of the parking lifecycle, processing all fee computations while the vehicle's physical location status securely remains 'PARKED'. The billing sequence initiates when a customer scans their original QR token at the exit scanner, prompting the backend database to calculate the exact elapsed duration. Based on this computation, the system dynamically routes the transaction through one of several strict billing scenarios: stays under 15 minutes trigger a free grace period (updating Payment_Status to 'BYPASSED'); standard daytime parking incurs a flat ₱30.00 fee ('PAID'); and vehicles remaining past 23:00 or rolling over into a new calendar day automatically trigger a ₱100.00 overnight fee ('PAID_OVERNIGHT'). For pre-planned stays, the system enforces a strict digital reservation pricing matrix: a base fee of ₱60.00 per day, mathematically stacking an additional ₱100.00 overnight surcharge plus the subsequent ₱60.00 daily fee for any bookings spanning two days or

crossing the overnight threshold. To support the facility's commercial ecosystem efficiently and eliminate redundant paper waste, the module handles vendor validations by allowing retail tenants to simply scan the customer's original entry QR code. This digital handshake directly updates the token's database status to 'PAID_VENDOR', granting the customer a strict 20-minute exit window. Finally, the system securely handles exceptions via a Lost Ticket protocol, where untracked customers are penalized with a ₱100.00 surcharge on top of their accrued fees. Ultimately, every state change generates a transaction receipt and a tamper-proof financial audit record, completely eliminating manual calculation errors and revenue leakage.

1.7.3.4 Exit Verification & State Finalization Module

EXIT VERIFICATION & STATE FINALIZATION MODULE

This module acts as the secure physical gatekeeper of the closed-loop architecture, executing a strict Dual-Token verification sequence that cross-references the scanned QR token's cleared payment status with YOLOv8-extracted license plate data to automatically actuate the exit barrier, and finalize the vehicle's database state.

 U2

 P1, P2

 OP1, OP2,
OP3

 F1, F2, F3, F4

 IN2, IN4

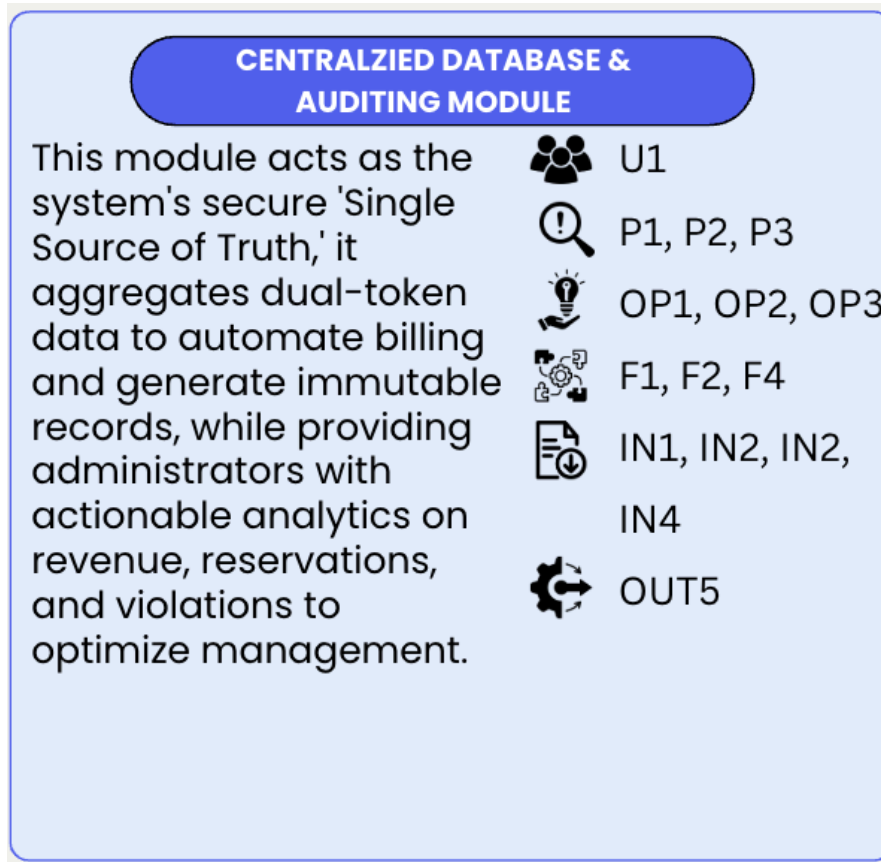
 OUT1, OUT5

Description:

This module acts as the facility's secure physical gatekeeper, enforcing the "Closed-Loop" architecture to prevent ticket-swapping fraud and unauthorized exits. The finalization sequence is

triggered when a customer approaches the exit and gets their license plate captured by the exit-facing CCTV CAMERA. In a case where the LPR fails, a customer will then scan their Quick Response (QR) token at the exit terminal as a fallback. Upon receiving these inputs, the Python backend queries the centralized SQLite database using the scanned QR ID to retrieve the vehicle's current Payment_Status and the original visual string (Logged_Plate) captured during entry. The system then executes a strict decision matrix: it verifies that the financial obligation is cleared (status equals 'PAID', 'BYPASSED', 'PAID_VENDOR', 'PAID_OVERNIGHT' or 'PAID_RESERVATION') and performs a comparative logic check to ensure the newly extracted Camera_Plate exactly matches the Logged_Plate. Crucially, to accommodate the physical limitations of motorcycles, the algorithm gracefully bypasses the visual check if the Logged_Plate is explicitly recorded as NULL, relying solely on the validated QR status. If the dual-token verification succeeds, the system updates the database Location_Status to 'EXITED', automatically signals the microcontroller relay to actuate the exit boom barrier, and asynchronously updates the spatial allocation module to reflect the freed parking slot. Conversely, if the transaction is 'UNPAID' or a plate mismatch occurs, the hardware relay maintains a closed state, effectively mitigating revenue leakage.

1.7.3.5 Centralized Database & Auditing Module



Description:

This module operates as the system's "Single Source of Truth," functioning as a highly secure, relational SQLite database that underpins all transactional and spatial logic within the facility. By continuously aggregating data streams from the entry and exit cameras alongside physical QR token scans and vendor validations, the module systematically enforces strict billing rules, such as grace periods and overnight thresholds, by calculating the exact duration of each parking session. Crucially, it inextricably links the finalized Payment_Status (e.g., standard, vendor-validated, or overnight) with the Dual-Token identifiers—the generated QR ID and the visual license plate string—to create an immutable audit record. This centralized architecture directly resolves the facility's legacy vulnerabilities by completely eliminating manual "notebook" recording errors,

closing the "Open Loop" physical security flaw, and rectifying the historical absence of entry auditing. Furthermore, by maintaining a persistent, real-time ledger of both physical vehicle occupancies and financial clearances, the module equips the facility administrator with irrefutable proof of daily revenue and precise, data-driven capacity trends, effectively transforming raw operational inputs into actionable business intelligence.

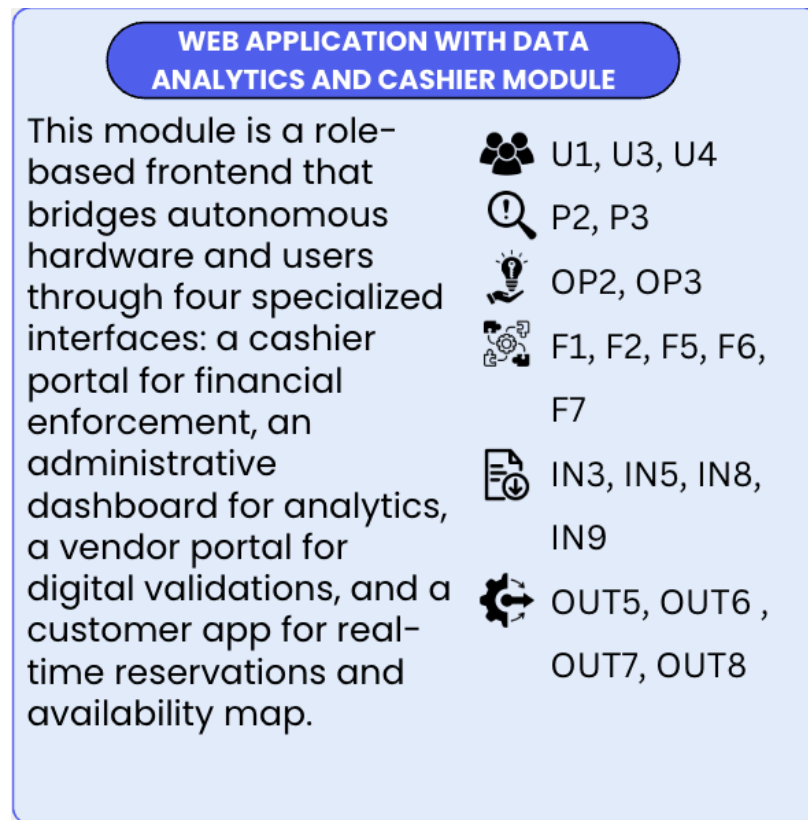
Building upon this centralized data architecture, the module extends its functionality into a comprehensive analytics and reporting layer that transforms raw transactional data into structured financial and operational insights. For the financial side, the system generates a Revenue Summary that consolidates total earnings across daily, weekly, and monthly periods, providing administrators with clear visibility into overall performance. Revenue streams are further segmented into reserved slot income, overnight parking charges, and regular parking fees, allowing for granular analysis of income sources. The module also generates reports that identify peak revenue hours, enabling operators to understand high demand periods, while calculating the average revenue per vehicle to assess pricing efficiency and utilization profitability. Complementing this, the system supports trend and comparative analysis, offering evaluations of daily, weekly, and monthly performance metrics. It distinguishes peak versus off-peak earnings, highlighting demand fluctuations, and computes growth rates to indicate increases or declines in revenue over time. These insights empower management to make data-driven adjustments to pricing strategies, staffing, and capacity planning.

In terms of reservations, the module provides a detailed Reservation Summary, tracking the total number of bookings across different timeframes. It categorizes reservations into active, completed, and no-show instances, enabling administrators to evaluate user reliability, optimize slot allocation, and minimize revenue losses caused by unused reservations. From an enforcement perspective, the system maintains a robust Violations Summary, recording all infractions on a daily, weekly, and

monthly basis. Violations are classified into specific types such as illegal parking, non-payment, and unauthorized slot usage, while also identifying repeat offenders through historical pattern tracking. This structured categorization enhances both operational enforcement and policy evaluation. Finally, the module includes a Penalty Summary that aggregates all financial penalties issued within the system. It tracks total fines imposed, collected payments, and outstanding balances, providing a clear view of enforcement effectiveness and revenue recovery. Additionally, the system computes the average fine amount, offering insights into penalty consistency and deterrence strength.

Together, these reporting capabilities elevate the module beyond a transactional database into a fully integrated decision-support system, enabling administrators to monitor financial health, enforce compliance, and continuously optimize parking operations with precision and accountability.

1.7.3.6 Web Application with Data Analytics and Cashier Module



Description:

This module operates as the system's secure, role-based frontend, bridging the autonomous hardware logic with human-in-the-loop oversight through three specialized user interfaces. Page 1 (Attendant Cashier Interface) empowers operational staff to manually process financial clearances and actively enforce facility rules. Attendants can query vehicle records via QR scans or direct license plate inputs to process standard cash or e-payments; crucially, this interface securely integrates the Lost Ticket protocol, automatically calculating accrued duration fees alongside the requisite penalty surcharge for un-ticketed vehicles. Upon clicking "Confirm Payment," the system rigidly logs the specific Staff_ID for accountability and updates the database state to 'PAID' to satisfy the exit barrier logic, while simultaneously functioning as the active monitoring hub that receives real-time Spatial Anomaly Alerts whenever the vision module detects unauthorized parking.

Page 2 (Administrative Analytics Dashboard) serves as the executive business intelligence hub for the facility owner, aggregating all transactional data to visualize revenue by payment type, tracking manual cashier logs to permanently eliminate "Open Loop" revenue leakage, and generating vision-backed Capacity Heatmaps to optimize staff deployment during peak congestion.

Page 3 (Tenant Vendor Portal) provides commercial partners with a streamlined, isolated interface; rather than dispensing physical drive-thru coupons, vendors simply utilize this page to scan a customer's original entry token, instantly updating the database state to 'PAID_VENDOR' and securely initiating the 20-minute exit window while completely eliminating redundant paper waste.

Finally, Page 4 (Customer Page) allows the customer to create an account to register their license plate into the system so that the LPR cameras' will recognize the plate number if they have a reservation. The web app will also be the primary way for the customer to reserve a slot. The customer will be able to select which slot they want to reserve, from Zone B-E, and for how long (1-3 days). Zone A will be reserved for accounts that are confirmed to be PWD citizens and any VIP personnel entered into the system by the administrator. Next, the web app pulls live data to display the real-time availability of parking slots to the user. Before even arriving at the facility, users can open the app to see exactly how many slots are open in Zones A, B, C, D, or E. Once a user enters the facility and the Vision System assigns them a slot, the app dynamically updates to guide the user to their designated space. Lastly, customers can check the web app if they have any outstanding bills for violations committed inside the parking lot. The customer can enter their license plate into the web app and it will show any violations that the customer has not paid yet.

1.8. Assumptions, Scope, and Limitations

This section defines the operational assumptions, system scope, and limitations of the proposed Automatic Unmanned Parking Management System, ensuring clarity in system boundaries and implementation feasibility.

1.8.1 Assumptions

The system development and deployment rely on the following assumptions:

Hardware Availability:

It is assumed that high-resolution cameras, stable network connectivity, and reliable power supply will be available to support continuous system operation.

Model Accuracy:

The YOLOv8-based License Plate Recognition (LPR) system, integrated with Tesseract OCR, is assumed to achieve sufficient accuracy in detecting and recognizing license plates under standard environmental conditions.

Standard Plate Usage:

All vehicles entering the facility are assumed to use officially issued and readable license plates compliant with Land Transportation Office (LTO) standards.

1.8.2 Scope

Process

The system is specifically tailored for Parking Ltd. in Cabanatuan City, based on our operational data gathered through stakeholder interviews and on-site observations.

The system covers the parking lifecycle from the moment a vehicle enters the facility, including identification and monitoring, up to the point where the vehicle completes payment and exits the premises.

Vehicle

The system processes Class 1 vehicles only, including: Sedans, SUVs, Vans, Pickup trucks, Motorcycles.

For motorcycles: Identification is limited to QR-based tracking only due to lack of front plates

License Plate

The LPR system is designed to recognize standard LTO-issued plate types, including:

- Private Vehicles: Black or green characters on white background or old Philippine license plate design
- Public Utility Vehicles (PUVs): Black characters on yellow background
- Government Vehicles: Red characters on white background
- Diplomatic Plates: Blue/black characters on specialized backgrounds
- Protocol Plates: Numeric-only plates assigned to government officials

The system focuses strictly on alphanumeric extraction from valid and visible plates.

1.8.3 Limitations

Environmental Constraints

Computer vision performance may degrade under:

- Heavy rain
- Low light conditions
- Plate obstruction or damage
- Extreme glare

Camera positioning and lighting calibration are critical to maintaining LPR reliability under such

environmental variations

Traffic Flow Sensitivity

Since parking Ltd is located adjacent to a highway, the system must maintain low processing latency. If detection time exceeds acceptable thresholds, temporary traffic buildup may occur. This means that system performance optimization is critical to prevent traffic.

Hardware Dependency

System reliability depends on:

- Camera quality
- Proper angle calibration
- Stable power supply
- Network connectivity

Hardware failure may temporarily require fallback manual override.

Human Intervention

Parking attendants act as system assisted operators rather than primary decision makers in slot allocation. They support real time operations by assisting with manual reservations, guiding drivers to assigned slots, and monitoring the system dashboard for availability and inconsistencies. They also handle manual payment processing for users without access to e-payment or those experiencing technical issues. Additionally, attendants manage exceptions such as system errors, reservation conflicts, and customer concerns, while enforcing parking policies and ensuring compliance

Motorcycle Processing Constraints

Due to standard vehicular design in the Philippines, motorcycles only have license plates mounted at the

rear. Consequently, the front-facing LPR cameras at the entry boom barrier cannot capture motorcycle plates upon approach. For motorcycles, the system's "Dual-Token Authentication" cannot be fully enforced; instead, motorcycle riders will rely exclusively on the generated QR code token for entry logging, duration calculation, and exit validation.

A camera capture demonstrating this limitation is shown in *Appendix E*

Parking Capacity

The system is designed to manage up to five designated parking sections within the facility. When all five sections reach full capacity, the system may no longer accommodate additional incoming vehicles within its standard allocation logic.

Although a sixth area, identified as a rough road section, exists, it is not implemented or integrated into the system's allocation, monitoring, and reporting mechanisms. As a result, this section is excluded from automated parking operations and may require manual handling for overflow vehicles.

A camera capture demonstrating this limitation is shown in *Appendix B*

Contextual Scope

The system is designed specifically for:

- Low speed private parking facilities
- Controlled entry and exit points

The system is not designed for:

- Large scale municipal smart city infrastructure
- Public law enforcement applications

1.9. Significance of the Study

The implementation of the Automatic Unmanned Parking Management System utilizing a dual-token approach (Quick Response codes and License Plate Recognition) holds substantial value for various stakeholders. The beneficiaries of this study are categorized into the direct operational users and the broader academic community.

1.9.1. Significance to the Target Beneficiaries of the Solution

1.9.1.1. Parking Facility Owner and Administrator

For the facility owner, this study provides a scalable, automated solution that eliminates the high costs and inaccuracies of manual monitoring. By shifting to a fully digital, unmanned dual-token system, the administration can definitively close the "Open Loop" security flaw, significantly reducing revenue leakage caused by unauthorized exits and unverified "pass-through" claims. Furthermore, the centralized SQL database provides the owner with accurate, real-time insights into facility utilization and tamper-proof financial audit logs.

1.9.1.2. Drivers and Customers

For transient parkers, students, and hospital visitors, the system replaces traditional queuing bottlenecks with a faster, streamlined, attendant-assisted checkout experience using a scanner. It reduces the stress and delays typically associated with manual payment booths and guard-dependent exit gates during peak hours (10:00 AM - 3:00 PM). The automated computation ensures that drivers are billed accurately and transparently based on exact timestamps, eliminating manual calculation errors.

1.9.1.3. Tenant Vendors (Commercial Establishments)

The system introduces a new layer of commercial synergy for the businesses operating inside the parking compound (e.g., Joey's Restaurant, 228 Cafe). It provides them with a structured digital framework to easily issue "PAID VIA VENDOR" QR codes. This empowers vendors to subsidize or validate parking for their short-stay retail and drive-thru customers seamlessly, boosting their foot traffic and sales without compromising the overarching security of the parking lot.

1.9.1.4. Parking Attendants

By automating the physical actuation of the boom barriers and the recording of license plates, the system relieves the existing staff from operational fatigue. Instead of being tied to a manual push-button and a handwritten logbook, attendants are freed from repetitive physical labor, allowing them to focus on higher-level facility maintenance, customer assistance, and exception handling.

1.9.2. Significance to the Academic Community

From a technical and academic perspective, this project contributes significantly to the fields of Information Technology and Computer Science by demonstrating the practical application of Deep Learning and the Internet of Things (IoT) in everyday commercial infrastructure. Specifically, it showcases how hybrid authentication models—pairing computer vision (YOLOv8 and Tesseract OCR) with dynamic digital tokens (QR codes)—can be engineered to handle complex, localized business rules, such as dynamic overnight fee thresholds and third-party vendor validation.

Furthermore, by utilizing the Scrum framework, as recommended by Sassa et al. (2023) and Verwijs and Russo (2022), the project serves as a real-world case study on ensuring high software quality

and team effectiveness through iterative development. As observed by Alami and Krancher (2022), documenting this methodology provides future researchers with a blueprint for ensuring that the integration between physical IoT hardware relays and complex software logic remains resilient and adaptable based on real-world performance data.

1.9.3. Significance to the Industry

The parking and urban mobility industry is currently undergoing a rapid digital transformation, shifting away from legacy, hardware-heavy infrastructure toward "ticketless" and frictionless operations. This study provides a highly relevant contribution to this transition. By demonstrating the viability of integrating off-the-shelf consumer hardware (generic CCTVs) with advanced computer vision (YOLOv8) and digital tokens (QR codes), the research presents a cost-effective blueprint for Small and Medium Enterprises (SMEs) to adopt Smart Parking technologies.

Industry reports emphasize that replacing physical tickets and manual enforcement with Automatic License Plate Recognition (ALPR) significantly reduces operational bottlenecks, minimizes paper waste, and prevents revenue leakage caused by inaccurate data collection. Furthermore, inefficient parking management is a primary driver of urban congestion, as drivers spend excessive time "cruising" for available slots. By automating occupancy tracking and integrating real-time capacity displays, systems like the one proposed in this study directly support broader urban mobility goals—reducing vehicle idling times, lowering carbon emissions, and creating a more seamless, sustainable transportation ecosystem. Ultimately, this research validates that enterprise-level parking automation is achievable and scalable for local, privately-owned facilities, driving the industry toward more accessible smart city innovations

CHAPTER 2. Review of Related Literature

2.1. Review of Related Concepts

2.1.1 Traditional Parking Management and Its Limitations

Parking management is a growing challenge in many urban environments as vehicle ownership rises. Conventional systems depend heavily on manual processes, such as physical ticketing, guard monitoring, and paper-based record keeping. These methods are not only time-consuming but also prone to human error, resulting in operational inefficiencies, increased wait times, and higher manpower costs. The inefficiencies become particularly apparent during peak hours, where manual systems struggle to handle the volume of vehicles efficiently, contributing to congestion and resource strain in facilities such as malls and public parking lots.

2.1.2 Smart and Automated Parking Systems

To address those limitations, researchers have investigated a range of smart parking technologies. These include sensor-based systems using ultrasonic, infrared, or magnetometer sensors to detect space occupancy. While these systems improve upon manual methods by automating parking space detection, they often involve significant infrastructure costs and may not scale well across large or open parking areas.

Vision-based and computer vision techniques have emerged as promising alternatives. Through the use of camera feeds and image processing, these systems can identify parking occupancy and other vehicle characteristics without relying on expensive or invasive physical sensors. Image processing makes use of techniques such as noise reduction, grayscale conversion, and object detection to interpret camera images, enhancing the accuracy and flexibility of smart parking solutions.

2.1.3 License Plate Recognition (LPR/ALPR) as an Enabling Technology

A core component of modern automated parking systems is Automatic License Plate Recognition (ALPR), also known as License Plate Recognition (LPR). ALPR systems use image processing and optical character recognition (OCR) to capture and translate visual license plate data into machine-readable text. This technology eliminates the need for manual plate entry and makes vehicle identification fast and accurate.

ALPR techniques typically include:

- Image acquisition through high-resolution cameras
- Pre-processing (e.g., grayscale conversion, noise reduction) to enhance readability
- Character recognition using OCR models to extract textual license data

These stages together make it possible to automate vehicle tracking, access control, and billing in parking management.

Studies have shown that ALPR systems can perform reliably even under real-world conditions, though methods such as advanced deep learning models (e.g., YOLO) further improve detection accuracy and robustness across varying lighting and environmental scenarios.

2.1.4 Integration of LPR in Parking Management

Several existing systems illustrate how license plate recognition directly contributes to parking automation. In an Intelligent Parking System (IPS) leveraging a YOLO object detection model, ALPR is used

to automatically detect and recognize vehicle plates upon entry without manual input, improving accuracy and reducing maintenance needs by eliminating extraneous sensors. Such systems also integrate modern payment methods (e.g., QR codes) to further streamline operations and user experience.

Other research demonstrates smart parking management that uses an image processing-based ANPR approach combined with IoT hardware. These systems automate vehicle registration and maximize parking space utilization while removing the need for manual record keeping.

Collectively, these studies show that ALPR can serve as a *central mechanism* for vehicle identification and control in automated parking — highlighting its potential as a foundation for *unmanned operations*.

2.1.5 Benefits and Performance Outcomes

Integrating image-based vehicle recognition into parking management has been shown to yield several advantages:

- Improved efficiency and throughput: Real-time plate recognition speeds up vehicle entry and exit, reducing queues.
- Reduced dependence on manpower: Automated systems minimize the need for guards and manual ticketing.
- Greater accuracy: OCR-based recognition reduces errors associated with manual license plate entry, thereby improving billing and record keeping.

These benefits directly support the core goals of your proposed system: improving efficiency, accuracy, and convenience in parking management through automation.

2.1.6 Gaps and Opportunities for Improvement

Despite progress, existing research also highlights ongoing challenges:

- Robustness to real-world conditions: ALPR performance can still be affected by lighting variations, occlusions, and plate diversity across regions.
- Integration and scalability: Not all systems seamlessly combine ALPR with complete parking management functions such as dynamic allocation, fee computation, and unmanned access control.

These gaps point to opportunities for new research — specifically, the development of a fully automatic unmanned parking management system that integrates image-based license plate recognition with parking entry/exit control and backend database functions, thereby offering a holistic and scalable solution.

2.1.7 Legal Implications and Responsibilities in Automated Data Collection

In the context of automated parking management, the deployment of License Plate Recognition (LPR) cameras and digital database logging introduces strict legal responsibilities. Because a vehicle's license plate number can ultimately be traced back to its registered owner, it is treated as personal data under Philippine law. Parking Ltd. must comply with national data protection and cybersecurity frameworks to ensure the integrity of its data collection process and protect against the risks of data breaches, surveillance abuse, and unauthorized profiling. These responsibilities are governed by the following laws

and guidelines:

2.1.7.1 Data Privacy Act of 2012 | Republic Act No. 10173

Under the Data Privacy Act (DPA), a license plate number is classified as personal information because it can reasonably identify an individual when combined with other records. Organizations like Parking Ltd., acting as Personal Information Controllers (PICs), are mandated to adhere to the core principles of transparency, legitimate purpose, and proportionality. The facility must inform users that their vehicle data is being recorded for the sole purpose of parking management, security, and billing calculation. The law strictly prohibits the use of this captured LPR data for unauthorized tracking, marketing, or third-party sharing without explicit consent. Employers and administrators can face severe penalties, including fines and imprisonment, for unauthorized processing or gross negligence in protecting this data (R.A. 10173, 2012).

2.1.7.2 NPC Advisory No. 2020-04 | Guidelines on the Use of CCTV Systems

Because the proposed system utilizes continuous entry and exit camera feeds paired with YOLOv8 to capture vehicle data, its physical infrastructure falls under the purview of the National Privacy Commission's (NPC) guidelines on CCTV usage. The advisory mandates that establishments must place prominent and highly visible notices at the entrance indicating that CCTV surveillance and LPR systems are in operation. This serves as "implied consent" from the driver upon entering the premises. Furthermore, Parking Ltd. is required to establish strict data retention policies, ensuring that video feeds and audit logs are securely deleted after they have fulfilled their designated purpose (e.g., after the parking fee is settled and a reasonable financial audit period has passed), thereby minimizing the risk of unauthorized long-term surveillance (National Privacy Commission, 2020).

2.1.7.3 Cybercrime Prevention Act of 2012 | Republic Act No. 10175

With the transition from physical handwritten notebooks to a Centralized Database System and a Web-based Management Portal, Parking Ltd. assumes the legal responsibility of securing this digital infrastructure. The Cybercrime Prevention Act penalizes illegal access, data interference, and computer-related fraud. It is the facility's legal obligation to implement reasonable cybersecurity measures—such as Role-Based Access Control (RBAC) to differentiate cashier permissions from administrative rights—to protect the stored transaction histories, generated QR tokens, and license plate logs. Negligence in securing this centralized database against malicious hackers or internal data leaks could lead to significant institutional liability and compromised customer safety (R.A. 10175, 2012).

2.2. Review of Related Systems / Infrastructure

2.2.1 IoT and Machine Learning Framework

Modern infrastructure for unmanned parking is increasingly moving towards a "Dynamic Parking Management" approach that aims to remove human intervention entirely. Alymani et al. (2025) propose a comprehensive framework that tightly integrates Internet of Things (IoT) sensors with machine learning algorithms to handle access control autonomously. Instead of relying on simple proximity sensors, this architecture utilizes a centralized processing unit that verifies vehicles against a cloud database in real-time. This seamless handshake between physical hardware and software logic allows the system to log entry times and calculate fees instantly, significantly streamlining the parking process. By leveraging this IoT-driven architecture, facilities can reduce the congestion and operational overhead typically associated with traditional manual systems.

2.2.2 YOLO and Tesseract OCR Recognition

The software infrastructure for recognizing vehicles has evolved from traditional image processing to advanced Deep Learning models that offer greater speed and precision. Tagalanao et al. (2024) demonstrated a cost-effective system architecture utilizing the YOLO (You Only Look Once) algorithm for vehicle detection paired with Tesseract OCR for character extraction. Their research highlights that consumer-grade webcams, when combined with efficient Python-based algorithms, can achieve high accuracy without the need for expensive industrial-grade sensors. This setup processes video feeds locally to classify parking slots and log license plates, proving that professional-level performance is achievable with accessible hardware. Such an approach is critical for developing effective unmanned systems that remain within a reasonable implementation budget.

Redmon et al. (2016) introduced the YOLO algorithm as a deep learning based object detection framework designed for real time detection tasks. The model operates as a single stage convolutional neural network that processes an entire image in a single pass, this allows objects within the image to be detected and classified simultaneously. Unlike traditional multi stage detection frameworks that require separate region proposal and classification stages, YOLO integrates these processes into a unified architecture. This design significantly improves processing speed while maintaining detection accuracy, making the algorithm highly suitable for applications requiring real time monitoring.

Recent improvements in the YOLO architecture have further enhanced its detection capabilities. Modern implementations like the YOLOv8 improve the detection of small objects, reduce inference latency, and optimize computational efficiency when processing live video streams. According to Satya et al. (2025), these improvements allow YOLO based systems to achieve faster and more reliable detection performance

in automated license plate recognition applications. Similarly, Park et al. (2022) demonstrated that YOLO based models can effectively detect vehicles and license plates within complex traffic environments, supporting their use in intelligent transportation and parking management systems.

Optical Character Recognition (OCR) technology is used to extract the alphanumeric characters from the plate image. Du et al. (2012) explained that OCR engines are essential components of automatic license plate recognition systems because they convert visual character data into machine readable text. One widely used OCR engine for this purpose is Tesseract OCR due to its open source availability and reliable text recognition performance when combined with image preprocessing techniques.

Before the character extraction process, the detected license plate image undergoes preprocessing steps such as grayscale conversion, thresholding, and noise reduction using computer vision tools like OpenCV. These preprocessing methods improve the contrast between the characters and background noise, allowing the OCR engine to accurately interpret the characters on the license plate.

2.2.3 Auto Gate Smart Parking (IoT)

To achieve a truly unmanned environment, the physical actuation of barriers must be triggered directly by recognition software rather than manual driver inputs. Kamaruddin et al. (2020) developed the "Auto Gate Smart Parking System," which modifies standard hardware infrastructure to replace RFID cards and ticket machines with digital authentication. In this configuration, the barrier gate receives open commands directly from the central processing unit once a license plate is successfully verified, eliminating the need for drivers to interact with kiosks. This infrastructure change not only accelerates the entry process but also reduces the mechanical wear and maintenance costs associated with physical ticket dispensers. Consequently, this system design offers a seamless "drive-through" experience that is essential for modern

automated facilities.

2.2.4 Robustness Against Environmental Factors

A critical challenge in the infrastructure of unmanned systems is maintaining reliability across varying environmental conditions such as lighting and weather. Mokayed et al. (2022) emphasize that standard camera setups often fail when faced with low light, rain, or oblique viewing angles, which leads to significant recognition errors. Their review suggests that robust infrastructure requires specific sensor positioning or the implementation of advanced processing techniques, like "Phase Congruency Models," to handle perspective distortion effectively. Without accounting for these physical variables, an automated system may suffer from downtime or require human intervention during adverse weather. Therefore, designing a resilient physical setup with adequate lighting and optimal camera angles is just as important as the software logic itself.

2.2.6 Real Time LED Parking Capacity Display

Real time parking information systems are increasingly used in intelligent parking infrastructures to inform drivers about available parking spaces before entering a facility. Providing drivers with real-time parking availability information helps reduce the time spent searching for parking spaces and improves traffic flow within parking facilities. According to Lin et al. (2017), smart parking systems that provide real-time occupancy information significantly improve parking efficiency by allowing drivers to determine parking availability before entering a parking area.

Digital display technologies such as LED panels are commonly used in parking facilities to communicate occupancy information to drivers. LED displays are particularly suitable for outdoor environments because

they provide high brightness and strong visibility. These displays can present real time parking availability information, such as number of available spaces or overall capacity percentage, allowing drivers to quickly determine whether parking spaces are available before entering the facility.

Luque-Vega et al. (2020) investigated an Internet of Things (IoT)–based smart parking architecture that monitors parking occupancy and communicates the collected data to centralized systems for real-time processing. Their study demonstrated that parking occupancy information gathered from monitoring technologies such as sensors or vision-based detection systems can be stored in a central database and transmitted to display interfaces installed at the entrance of parking facilities. These findings suggest that integrating database-driven monitoring with digital display systems enables parking infrastructures to provide real-time visual feedback to drivers, improving parking management efficiency and reducing unnecessary vehicle circulation within parking areas.

2.2.7 SQLite Database Management System

Database management systems play a crucial role in software applications that require reliable storage and retrieval of structured data. In automated parking management systems, databases are used to store vehicle information, license plate records, timestamps, and system logs generated during operation. Selecting an appropriate database is essential to ensure efficient data processing while maintaining system simplicity and stability.

SQLite is a lightweight, serverless relational database engine widely used in embedded systems and small to medium scale applications. Unlike traditional database management systems that require a separate database server, SQLite operates as a file based database that runs directly within the application environment. Pawlaszczyk (2022) explains that SQLite is designed to be self-contained, requiring minimal

configuration while providing reliable data storage through a structured relational model. Because of this architecture, SQLite is particularly suitable for systems that operate locally and do not require high levels of concurrent database access.

Another advantage of SQLite is its efficiency in environments with low to moderate transaction loads. Ezginci and Karadaş (2019) investigated the performance of SQLite in real-time access control systems and found that it performs effectively when handling localized data storage and limited concurrent requests. Their findings suggest that SQLite can support applications where data is generated continuously but accessed by a small number of processes, making it appropriate for embedded monitoring systems and automated entry control systems.

2.2.8 Quick Response (QR) Code Technology in Smart Parking

Quick Response (QR) codes have become a ubiquitous tool in modern digital infrastructure due to their low implementation cost, high data storage capacity, and rapid readability. In the context of smart parking, QR codes serve as a dynamic and cost-effective alternative to traditional hardware-based tokens like RFID tags or magnetic strip tickets.

According to Patil and Patil (2019), implementing a QR code-based framework significantly reduces the capital investment required for parking facilities by eliminating the need for expensive physical sensors and automated ticket dispensers. Their research highlights that utilizing dynamically generated QR codes for vehicle identification streamlines the entry process and provides a seamless, touchless user experience. By scanning the code, the system securely logs the exact entry time directly into a centralized database. This acts as a unique digital identifier that facilitates highly accurate and automated billing upon

exit, all while reducing the mechanical wear typically associated with legacy hardware.

Furthermore, QR codes offer significant flexibility in handling complex commercial workflows. Zakaria et al. (2025) emphasize that when QR technology is paired with Automatic License Plate Recognition (ALPR), it creates a highly resilient "dual-token" authentication system. While the ALPR handles physical vehicle tracking, the QR code manages the financial state and dynamic billing rules. This separation allows external stakeholders—such as retail tenants or drive-thru vendors operating within a parking compound—to seamlessly issue temporary QR bypass coupons. When scanned, these specific QR tokens trigger localized business logic in the database, granting the user a predefined grace period or fee exemption without compromising the overarching security of the parking facility.

2.3. Review of Related Methodologies

2.3.1 Scrum Framework

The Scrum framework is a widely adopted agile methodology that emphasizes iterative development, collaboration, and adaptability. Recent empirical studies have formalized theories regarding how Scrum drives both team effectiveness and software quality, which are critical for the development of complex systems.

Verwijs and Russo (2023) propose a "Theory of Scrum Team Effectiveness" based on a large-scale study of approximately 2,000 Scrum teams. Their research identifies five high-level factors that determine a team's effectiveness: responsiveness, stakeholder concern, continuous improvement, team autonomy, and management support. A key finding of their model is that "responsiveness" defined as the capability to respond quickly to changing needs through frequent releases and backlog refinement is positively related with both stakeholder satisfaction and team morale. However, the study notes that responsiveness alone is

insufficient; it must be paired with "stakeholder concern," which ensures the team possesses a shared understanding of what is valuable to the stakeholders. This means that in order for a project to be successful, the development team must collaborate actively with stakeholders to verify the value of each iteration in addition to developing quickly.

Alami and Krancher (2022) look into how Scrum helps achieve software quality. They believe that Scrum enhances quality through "process-induced advantages" like iterative development as well as specific "social antecedents" like teamwork, psychological safety, accountability, and transparency. For instance, the study emphasizes that "voluntary inspections," in which team members freely examine one another's deliverables, are made possible by transparency in Scrum teams and result in early error discovery and rectification. Additionally, Alami and Krancher discovered that "external quality" and "internal quality" are two different ways that Scrum practitioners perceive quality. According to their findings, Scrum fosters an environment that is naturally suited to producing high-quality software, hence acting as a supplement to existing quality assurance procedures.

2.3.2 Kanban Methodology

The Kanban Methodology is a lean concept adapted for software development which emphasizes visualizing workflows, limiting work-in-progress, and developing continuous, iterative improvement.

Kirovska and Koceski (2015) present the usage of Kanban in software development, detailing its foundation on five core principles commonly observed in successful organizational adoptions. Their research identifies these primary drivers as visualizing the workflow, limiting Work-In-Progress (WIP) using a virtual system, managing flow, making explicit management policies, and utilizing scientific models to improve collaboratively. A key mechanism of this methodology is the "virtual Kanban board," which

fundamentally shifts the development process from a traditional "push" system to a "pull" system. This ensures team members handle only one task at a time, pulling new work forward only after the current item is finished, thereby actively identifying bottlenecks and preventing workflow overload.

Furthermore, Kirovska and Koceski investigate the tangible benefits of Kanban through the empirical evaluation of a web-based application, KanbanMAK, across multiple medium-sized software projects. They determined that utilizing Kanban yields significant enhancements in the daily development process, specifically citing better flexibility, improved cooperation, and reduced time for completing working tasks. Most notably, the study highlights a 23% decrease in the discrepancy between estimated and real task completion times, demonstrating that Kanban drives better predictability and estimation capabilities. Ultimately, their findings emphasize that Kanban acts as an evolutionary method; rather than enforcing a completely new management framework, it applies specific metrics and visualization to identify inefficiencies and gradually improve an organization's existing development process.

2.3.3 Lean Methodology

The Lean methodology is a project management and process improvement framework that focuses on optimizing resources, eliminating waste, and efficiently delivering customer value. Yadav, Mittal, and Jain (2019) examine the application of Lean specifically within software development projects, observing that the methodology is driven by the need to deliver quality codes with new features at a reduced cost. Their research categorizes Lean implementation into three distinct levels: philosophy, principles, and practices. At the principles level, they identify five core drivers for successful adoption: identifying value, mapping the value stream, creating flow, establishing pull, and seeking perfection. The study highlights that inefficiencies in software development frequently stem from partial work done, extra unnecessary features, and time wasted during late-stage testing and debugging. By adopting Lean principles, organizations can actively

target and eliminate these specific root causes of waste.

CHAPTER 3. Methodology

3.1. Description

The study is a developmental research design utilizing a system development methodology to design, implement, and evaluate an Automatic unmanned parking system using image based license plate recognition technology. Developmental research is very appropriate for the study as it aims to create, test, and refine technological solutions to real world operational problems.

The research adopted an Agile development approach, more specifically the Scrum approach to help guide the iterative development of the system. Scrum specifically helps give stability to projects that involve both hardware and software integration, especially those that require continuous testing, improvement and refinements.

The study followed a design-and-build process, consisting of requirements analysis, architectural design, system development, hardware integration, system testing, performance evaluation and performance evaluation.

For validation of systems, the study employed experimental testing procedures to measure recognition accuracy, processing time, and transaction throughput. Performance metrics included detection accuracy rate, Optical Character Recognition (OCR) extraction accuracy, system response time, and successful database verification rate.

Data gathered from system testing were analyzed using descriptive statistical methods, including computation of mean processing time and percentage accuracy. The evaluation focused on determining whether the system meets the operational objective of reducing vehicle transaction time to under three seconds while maintaining high data integrity. The performance metrics were taken from multiple interviews conducted across a two week interview and testing period.

3.2. Description of Each Phases

Phase 1: Requirements gathering and planning

This initial phase involved identifying and documenting the system requirements based on the problem statement, objectives, and scope defined in Chapter 1.

Phase 2: System Architecture/Design

During this phase, the system's overall tiered architecture was designed and finalized following the Scan, Pay, Exit operational flow, which defines the sequence of vehicle detection, payment processing and exit authorization

Phase 3: System Development and Integration

The system development was carried out incrementally using Scrum sprints. Where each sprint focused on completing specific functional modules. Software components such as license plate detection, Optical Character Recognition (OCR) processing, database management, and gate control developed and integrated with current hardware

Phase 4: System Tests

After full system integration, controlled testing is conducted to verify that the system works as intended. This phase ensures that both hardware and software components functions correctly in a unified system

Phase 5: Evaluation

Results from system testing will be analyzed using descriptive statistical methods to find any problems, bugs, errors or inefficiencies.

Phase 6: Refinement

Based on the results of the evaluation phase, identified errors such as recognition errors, system latency, or integration problems will be addressed through iterative refinements so that the system will be optimized to achieve any targets set.

3.2.1 Technologies Used per Development Phase

Phase 1: Requirements gathering and planning

This phase focused on identifying the system requirements, operational objectives, and system constraints based on the problem defined in Chapter 1. Technologies and Tools

Used:

- Documentation and planning tools for requirement specification
- System modeling tools for conceptual workflow development
- Basic hardware feasibility assessment for cameras, sensors, and barrier gate components.

These tools were used to establish the functional requirements of the automated parking system, including vehicle detection, license plate recognition, payment verification, and barrier gate automation.

Phase 2: System Architecture/Design

This phase involved designing the system's architecture and defining the interaction between hardware and software components. The system followed a Scan–Pay–Exit workflow architecture. Technologies Used:

- Python for backend system logic and model integration
- YOLOv8 for real-time license plate detection
- Tesseract OCR for extracting alphanumeric characters from detected plates
- OpenCV for image processing and camera input handling
- SQLite for database design and data storage.

During this phase, the interaction between the detection system, recognition engine, database verification, and barrier control system was structured.

Phase 3: System Development and Integration

In this phase, the system modules were developed incrementally using Scrum sprints. Software components were coded and integrated with the hardware infrastructure of the automated parking system.

Technologies Used:

Software:

- Python for system logic and API integration

- YOLOv8 for vehicle license plate detection
- Tesseract OCR for text recognition from license plates
- OpenCV for real-time camera processing
- SQLite for transaction and vehicle data storage

Hardware:

- IP or USB camera modules for capturing vehicle images
- Microcontroller-based relay modules for barrier gate control
- Barrier gate motor system for automated entry and exit control

The integration ensured that vehicle detection, OCR extraction, database verification, and barrier control functioned as a unified system.

Phase 4: System Tests

This phase involved controlled testing of the system to evaluate recognition accuracy and system response time.

Technologies Used:

- Python for running testing scripts and measuring performance metrics
- OpenCV for capturing test images during recognition trials
- SQLite for recording transaction results
- Stopwatch to measure system response time

Testing scenarios included different lighting conditions, plate positions, vehicle approach speeds, and vehicle positions.

Phase 5: Evaluation

In this phase, the results from system testing were analyzed to determine whether the system met the required performance targets.

Technologies Used:

- Microsoft Excel for organizing and computing performance data
- Python for statistical calculations and result processing

Evaluation metrics included:

- License plate detection accuracy rate
- OCR character extraction accuracy
- Mean system response timeSuccessful database verification rate

Phase 6:Refinement

Based on evaluation results, adjustments were made to improve system performance and reliability.

Technologies Used:

- Python for algorithm optimization
- YOLOv8 retraining and parameter tuning
- OpenCV for improving image preprocessing
- SQLite for database query optimization

Refinements focused on reducing processing latency, improving OCR accuracy, ensuring reliable barrier gate activation,and improving penalty accuracy.

3.3. Work Plans

3.3.1 CAP1 80%

The first phase of the project allocates 80% of the overall developmental workload to Capstone 1. This is necessitated by the project's "Design-and-Build" nature, which requires the complete construction of the system's functional infrastructure before testing can commence.

This phase encompasses Phase 1 (Requirements Gathering), Phase 2 (System Architecture/Design) and the majority of Phase 3 (System Development and Integration) of the methodology.

The primary objective during this stage is to achieve a minimum viable product (MVP) in which a vehicle can successfully complete a full parking transaction within the system. This involves development and integration of key technical components: Implementation and training of the YOLOv8 model for license plate detection, Integration of Tesseract OCR for character extraction, Development of a centralized SQLite database to serve as the system's primary data repository, Engineering and integration of hardware relay controls to automate the parking barrier gates.

By the end of Capstone 1 the system should be capable of executing a complete end-to-end transaction, beginning with vehicle detection at entry and ending with exit authorization. This ensures that the MVP is fully operational and ready for systematic evaluation.

3.3.2 CAP2 20%

The remaining 20% allocated to Capstone 2, which focuses on the quality assurance and optimization stages outlined in Phase 4 (System Testing), Phase 5 (Evaluation), and Phase 6 (System Refinement).

With the physical and digital architecture already constructed, this phase shifts focus from system

building to performance tuning and validation.

The primary objective of this stage is to validate the system's performance against the defined operational targets, particularly the requirement that the vehicle transaction processing time is reduced to under three seconds. Additional testing will evaluate the robustness of the recognition system under varying environmental conditions such as: changes in lighting conditions, partial license plate obstruction, and variations in vehicle positioning

Through iterative refinements and system adjustments, this last phase ensures that the final system presented during the final defense is not only functional but also statistically validated and optimized for real world implementation.

3.3.3 Gantt Chart

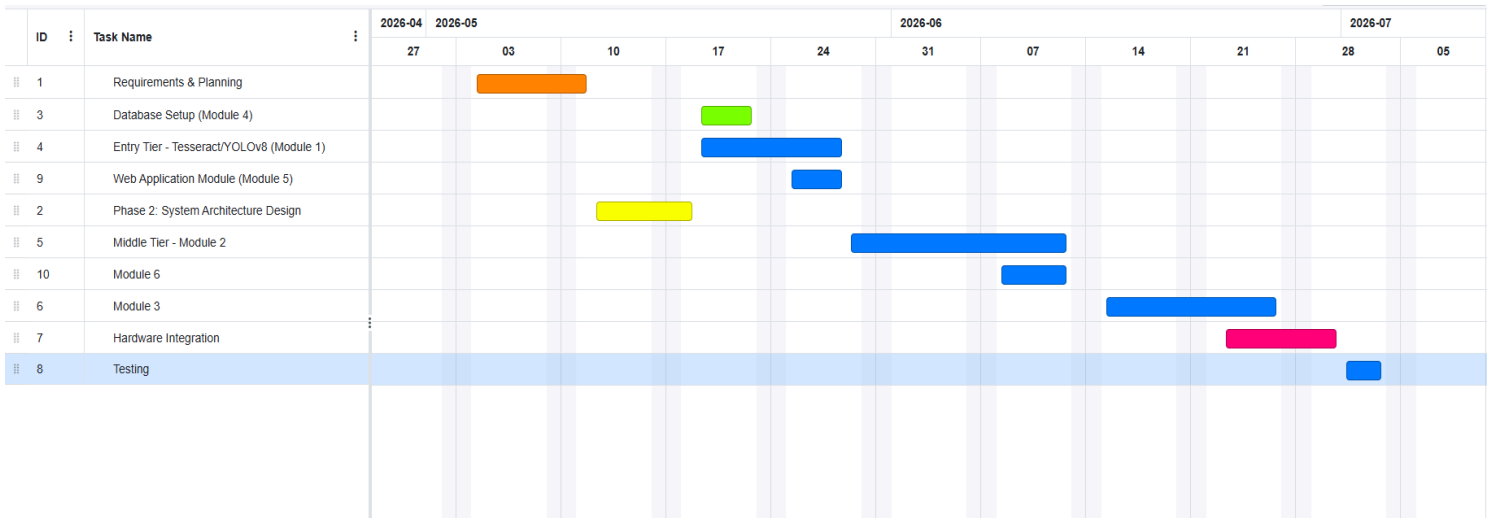


Figure 3.1 Gantt Chart of CAP1 Workload

The development stage is divided into several focused modules to ensure systematic progress. Initial setup includes database configuration and environment preparation, providing the foundation for

subsequent modules. The early tier module integrates core input functionalities such as OCR and camera processing, while the middle and exit tiers implement the system's decision logic, including paying at an exit scanner and barrier control mechanisms. Hardware integration is conducted afterward to connect relays and sensors with the software components, ensuring proper communication between physical and digital layers.

The project concludes with full system integration and testing, where all modules are combined and validated under real-world conditions. This phased approach promotes organized development, parallel task execution where feasible, and sufficient time for validation and refinement before final completion.

CHAPTER 4. Existing Systems

4.1. The Existing System

4.1.1 Organization

Parking Ltd is a privately owned, commercial open-air parking facility situated in a high-density urban district in Cabanatuan City. The facility is strategically located directly across from Doctor's Hospital and is surrounded by major educational institutions, including the College of the Immaculate Conception (CIC) and Our Lady of Fatima University (OLFU) Cabanatuan Campus (Loreto, personal communication, February 15, 2026). The facility operates under a sole proprietorship owned and managed by Mr. Roro James Cruz, who oversees general business strategy and revenue auditing. Daily ground operations are run by two rotating parking attendants—Jomar and Loreto—who handle manual barrier operations, cash collection, physical logbook entries, and custodial tasks. Additionally, the facility grounds house commercial tenants, including Joey's Restaurant, 228 Cafe, Chick Chicken, and Detalye Car Detailing Services, which operate under administrative agreements with the proprietor.

4.2. Description of the Existing System

4.2.1 Entry, Exit, and Access Control Process

The facility operates on a manual, attendant driven access workflow across an open air lot comprising both paved sections and unpaved rough terrain. The entry and exit points rely on basic mechanical barriers operated without digital integration or sensor driven logic. When a vehicle arrives, an attendant must manually trigger the barrier, inspect the vehicle, and record entry details on paper coupons or in a physical logbook.

Upon exit, drivers present their paper coupons to the attendant, who calculates the duration of the stay manually or applies a flat rate. Once payment is handed over in cash, the attendant opens the barrier to allow the vehicle to leave.

4.2.2 Space Allocation and Capacity Management

Parking Ltd has a maximum physical capacity of approximately 100 vehicles. Because the facility lacks an automated guidance or occupancy tracking system, parking spot selection relies entirely on "blind entry." Drivers enter the lot without knowing if spaces are available, cruising through the facility to locate an open spot. During high-density daytime hours (10:00 AM to 3:00 PM), when hospital visitors, students, and university faculty compete for limited space, this lack of spatial management leads to internal congestion, bottlenecks at the gate, and frequent instances of double parking.

4.2.3 Fee Structure and Validation Workflow

The facility imposes a flat fee structure of ₱30.00 for day parking, with an overnight fee of ₱100.00 applied to vehicles remaining past 11:00 PM. Transaction records and fee verifications are processed entirely in cash by the active parking attendant. While commercial tenants leased within the perimeter (e.g., local cafes and detailing shops) serve parking patrons, there is no standardized, integrated system to issue digital tenant validations or discount credits. Any customer discounts or validation arrangements require manual verification or physical paper tokens provided by the vendors.

4.2.4 Data Logging and Monitoring Practices

All operational and financial metrics at Parking Ltd are maintained through manual paper-based logging. Attendants write down arrival times, entry numbers, and collected payments in physical logbooks. At the end of a shift or operating cycle, revenue auditing relies on matching the cash collected in the till against the handwritten log entries and physical paper ticket stubs. No centralized digital system exists to provide real-time occupancy counts, automated billing calculations, or digital audit trails.

4.2.5 Current State of Technology and Infrastructure

The existing operational setup at Parking Ltd is almost entirely analog:

- Access Control: Standalone electromechanical barriers without computer integration, loop detectors, or computer vision devices.
- Monitoring Capabilities: Zero real-time visibility into current occupancy levels, capacity limits, or spatial distribution within the lot.
- Security & Validation: Absence of License Plate Recognition (LPR) or digital tokening (e.g., QR codes) to match entering vehicles with exiting vehicles.

- Tenant Integration: Commercial tenants lack dedicated digital terminals or point-of-sale integration to issue dynamic parking validations

4.3.Problem Areas

An evaluation of Parking Ltd's existing system reveals operational difficulties grouped into Manpower, Materials, and Methods:

Manpower

- Task Overload and Attendant Fatigue: The two rotating attendants (Jomar and Loreto) are responsible not only for gate access, ticketing, and cash collection, but also for custodial and facility maintenance duties (e.g., restroom cleaning).
- Security Risks During Absence: When an attendant leaves the gate unattended to perform maintenance, the electromechanical barrier is often left open or unmonitored, creating vulnerabilities for unauthorized vehicle entries/exits and potential unrecorded exits.
- Human Error in Transactions: Manual record-keeping and ticket processing increase the risk of miscalculated fees, lost physical stubs, and inconsistent logging during peak hours.

Materials

- Lack of Automation Hardware: The lot lacks License Plate Recognition (LPR) cameras, QR scanner terminals, optical sensors, or automated gate controllers.
- Absence of Real-Time Display Infrastructure: No digital signage exists outside or inside the lot to notify incoming drivers of available spaces, forcing vehicles to enter blindly.
- Physical Lot Limitations: The division between paved sections and unpaved rough road sections makes physical slot marking difficult, worsening unguided and disorganized parking behavior.

Methods

- Reactive & Paper-Based Record-Keeping: Relying on physical stubs and handwritten logbooks makes real-time data analysis impossible and creates severe vulnerabilities to revenue leakage.
- Unstructured Spatial Allocation: Without a section-based slot assignment workflow, drivers double-park or park haphazardly, blocking driveways and delaying traffic flow.
- Fragmented Tenant Validation: Commercial tenants operating within the lot have no automated mechanism to validate customer parking, requiring manual paper-coupon exchanges that are vulnerable to loss or misuse.

4.4. Opportunities

An analysis of Parking Ltd's operational bottlenecks highlights significant opportunities for systematic modernization through the implementation of an Automatic Unmanned Parking Management System using Image-Based License Plate Recognition (LPR) integrated with Quick Response (QR) Code Technology.

- **Automated Access Control and Closed-Loop Security:** Integrating computer vision LPR cameras with dynamically generated QR tokens establishes a secure, closed-loop identity for every vehicle entering the lot. Capturing license plates at entry and pairing them with QR tokens automates barrier operations, eliminates ticket loss disputes, and ensures no vehicle can exit without verified payment.
- **Optimized Spatial Allocation and Guided Parking:** Replacing "blind entry" with a real-time spatial allocation engine allows the system to direct drivers to available sections based on live occupancy data. This reduces internal traffic congestion, prevents double parking, and eliminates bottlenecks during peak university and hospital hours.
- **Centralized Digital Management and Revenue Protection:** Implementing a web-based management dashboard replaces paper logbooks with automated, audit-ready digital records. The system dynamically applies complex business logic—such as daytime flat rates, overnight surcharges, and grace periods—eliminating manual calculation errors and preventing revenue leakage.
- **Tenant Integration and Modern Payment Workflows:** Deploying web-based validation tools allows commercial tenants (e.g., Joey's Restaurant, 228 Cafe) to validate parking digitally through thermal-printed drive-thru QR codes or vendor apps. Furthermore, offering online reservations and supporting both digital and attendant-assisted cash payments creates a versatile experience for all customer demographics.

CHAPTER 5. The Proposed System

5.1. System Description

The Automated Unmanned Parking Management System is a fully integrated, multi-tiered digital architecture designed to automate vehicle monitoring, space allocation, payment processing, and enforcement operations for Parking Ltd. The platform replaces the legacy, manual operational model—characterized by open-barrier entry, manual logbooks, blind capacity estimation, and high risk of revenue leakage—with a secure, data-driven framework. By integrating localized computer vision models (YOLOv8 and Tesseract OCR) with hardware automation (radar sensors, electromechanical boom barriers) and a central SQLite database, the system closes the operational loop from vehicle entry to exit.

The system comprises three primary operational layers tailored to physical and digital stakeholders:

- **Physical & Computer Vision Edge Infrastructure:** Positioned at the entry/exit points and overhead sectional zones (Sections A through E), high-resolution CCTV cameras capture live video feeds. A local processing PC ingests these feeds to perform real-time vehicle detection, Automatic License Plate Recognition (ALPR), and vehicle color identification. Interfaced radar sensors detect vehicle approach and clearance, triggering barrier actuations automatically upon digital verification.
- **Central Backend & Automated Logic Engine:** Driven by an SQLite Central Database, this layer serves as the operational brain. It executes a rule-based spatial allocation engine (routing vehicles systematically across Sections A through E based on reservations, PWD/VIP status, and walk-in priority), enforces dynamic tariff rules (standard ₱30.00 flat rate, 10-minute free grace period, and 11:00 PM overnight ₱100.00 surcharges), and manages system state transitions.
- **Role-Based Web Application:** A unified web platform provides dedicated portals for various

stakeholders:

- Administrator Dashboard: Offers executive oversight, historical reporting, sales analytics, and audit trailing.
- Parking Attendant Portal: Receives automated violation alerts with visual photo proof and provides controls for manual payment overrides during edge cases.
- Vendor Portal: Enables accredited tenant establishments to validate customer transactions and apply legitimate parking fee exemptions.
- Customer Portal: Allows drivers to create accounts, check real-time sectional slot availability, pre-book reservations, scan unique QR codes, and settle pending fees or violation fines.

5.2 System Objectives

5.2.1 General Objective

To design and develop a parking management system for Parking Ltd. that addresses inefficiencies in vehicle monitoring, ticket validation, billing accuracy, revenue control, and exit verification in the current parking operations.

5.2.2 Specific Objectives

- To reduce entry and exit processing delays by replacing manual ticket handling with a more efficient digital identification system.
- To enhance parking security by implementing a reliable vehicle verification mechanism that ensures proper vehicle-to-ticket matching.
- To ensure accurate computation of parking fees based on established business rules, including grace periods and overnight charges.
- To minimize revenue loss and prevent unauthorized exits through real-time payment validation prior to vehicle release.
- To improve vendor validation procedures by enabling accredited establishments to authorize legitimate parking fee exemptions.
- To strengthen transaction monitoring and reporting through centralized digital record management.

5.3 Technical Requirements

The proposed system requires a combination of hardware, computer vision frameworks, database systems, and networking components to fulfill its automated operational workflow:

5.3.1 Hardware Requirements

Edge Computer Vision Cameras: High-resolution CCTV cameras strategically deployed at entrance/exit gates and mounted overhead across paved Sections A, B, C, D, and E to continuously stream live video for object detection and Automatic License Plate Recognition (ALPR).

Proximity Sensors & Actuation Relays: Dual radar sensor loops installed at entry/exit lanes to detect vehicle presence and clearance, interfaced with microcontroller relays to automatically actuate electromechanical boom barriers.

On-Site Edge Processing PC: Dedicated local computing workstation with GPU acceleration (e.g., NVIDIA CUDA-enabled) deployed at the facility gatehouse to process live video feeds, execute computer vision models with sub-second latency, and ensure fast gate actuation to prevent highway traffic buildup.

Token Dispenser & Scanners: Physical/digital QR code printing and scanning terminals installed at entry and exit gates to issue tokens and perform fallback scan handshakes.

5.3.2 Software & Computer Vision Requirements

Object Detection Engine: YOLOv8 framework fine-tuned to isolate vehicle bounding boxes, perform real-time sectional occupancy counting, monitor spatial violations in "No Parking" zones, and detect license plate regions.

Optical Character Recognition (OCR): Tesseract OCR combined with OpenCV image preprocessing to extract alphanumeric license plate text and identify vehicle colors from cropped plate feeds.

Central Database Management System (DBMS): Lightweight, embedded SQLite relational database serving as the facility's "Single Source of Truth." It maintains real-time records for Location_Status, Payment_Status, entry/exit timestamps, vehicle metadata, and transaction logs.

Backend & Automation Logic: Python-based processing pipeline connecting computer vision outputs, hardware relay signals, and database triggers.

5.3.3 Web Application & Interface Framework

Responsive Multi-Portal Frontend: Web framework rendering four specialized user interfaces:

- Page 1 (Attendant Cashier Interface): Manual QR/plate lookups, cash collection logging, Lost Ticket processing, and real-time Spatial Anomaly Alert notifications.
- Page 2 (Administrative Analytics Dashboard): Executive business intelligence metrics for owner Roro James Cruz, displaying revenue breakdowns, manual cashier overrides, and vision-backed Capacity Heatmaps.
- Page 3 (Tenant Vendor Portal): Isolated QR scanning interface for commercial tenants (e.g., Joey's Restaurant) to grant digital PAID_VENDOR exemptions.
- Page 4 (Customer Web App): Driver portal for user registration, real-time slot availability viewing, advance slot booking (1–3 days), and online violation fee settlement.

5.4 Non Technical Requirements

The effective implementation and continuous operation of the system rely on operational, security, user-level, and compliance standards:

5.4.1 Operational & Performance Standards

Low Processing Latency: Sub-second ALPR processing and barrier actuation times at access gates to ensure smooth vehicle flow and prevent traffic spillover onto the adjacent main highway. High System Availability: Continuous 24/7 uptime during operating hours, supported by local edge computing that operates independently of cloud internet connectivity for gate processing.

Fallback Human-in-the-Loop Procedures: Defined operational protocols for parking attendants to handle system exceptions (e.g., severely obscured plates, heavy rain degradation, manual reservation overrides, cash handling, and Lost Ticket processing).

5.4.2 Security, Integrity, and Auditability

Role-Based Access Control (RBAC): Strict authentication and permission levels enforcing data isolation between Administrators, Cashier Attendants, Tenant Vendors, and Customers.

Immutable Transactional Auditing: Automated logging of all cashier actions, manual gate overrides, vendor validations, and system exceptions tied to specific Staff_ID records to eliminate revenue leakage and unauthorized access.

Closed-Loop Verification Protocol: Enforced physical security where exit barriers remain locked down unless both financial clearance (Payment_Status) and dual-token identity matching (Camera_Plate == Logged_Plate) are verified.

5.4.3 Usability and Ergonomics

Intuitive Visual Interface: Clear visual indicator maps on the customer web app and attendant portal showing live slot occupancy across Zones A–E and redirection prompts for Section F overflow.

Simplified Customer Onboarding: Straightforward web registration requiring minimal input fields (email, password, license plate) to enable advance reservations and billing checks.

Automated Anomaly Alerting: Pushed real-time visual and auditory dashboard notifications with attached photo proof when spatial parking infractions occur, allowing prompt attendant response without continuous manual monitoring.

5.5 Modules

5.5.1 Automated Entry & Spatial Allocation Module

This module serves as the facility's primary gateway, initializing the vehicle's digital lifecycle and actively mitigating internal congestion through intelligent spatial allocation. Upon a vehicle's arrival, the Python-based backend queries the Sectional Occupancy Module's YOLOv8 camera data to dynamically assign the driver to the most vacant paved sector (Sections A-E), or defaults to the unmonitored rough road (Section F) if the main facility is at maximum capacity. As a vehicle approaches the entry point, the radar sensor detects its presence and the system performs a Dual-Token authentication handshake: it dispenses a Quick Response (QR) token explicitly printed with the assigned parking section, while simultaneously utilizing the entry CCTV camera, OpenCV preprocessing, and Tesseract OCR to extract the vehicle's alphanumeric license plate. Finally, these two tokens—the QR ID and the visual plate string—are securely bound together in a centralized SQLite database alongside the exact entry timestamp and assigned section, instantly triggering an edge microcontroller relay to actuate the entrance boom barrier and allowing the driver to proceed directly to their designated zone while permanently resolving the facility's entry data deficit.

5.5.2 Sectional Occupancy, Reservation & Vision Enforcement Module

This module operates as the facility's dynamic spatial engine and active compliance monitor, continuously synchronizing real-time physical camera data with strict digital reservation business rules to rigidly enforce space allocation across six distinct parking zones. The reservation lifecycle is initiated strictly through the client-side web application, where a customer selects an available physical slot and inputs their specific alphanumeric license plate for a duration of one to three days. To optimize slot turnover and prevent monopolization, the system enforces rigid booking parameters: reservations are restricted exclusively to the current week, single user accounts are systematically limited to reserving one slot for one designated vehicle, and all transactions are strictly non-refundable upon payment. Once the digital fee is cleared, the database inextricably binds a 'RESERVED' status to both the specific license plate and the chosen slot, instantly generating a virtual QR token for the user.

By monitoring the paved Sections A through E using YOLOv8 object detection, the system calculates "True Availability"—mathematically deducting both physically parked vehicles and these active prepaid reservations from the maximum sectoral capacity. When a reserved customer arrives, the system prioritizes a frictionless License Plate Recognition (LPR) match via the entry camera to instantly actuate the barrier, while providing the virtual QR token as a robust fallback to bypass redundant physical token dispensation during environmental camera failures.

Beyond capacity calculations, the module's vision architecture actively enforces spatial discipline and reservation integrity. It continuously cross-references physical vehicle bounding boxes and extracted plates against their database-assigned sections. If the system detects a license plate mismatch at a specific reserved slot, or identifies a vehicle persisting within virtual "No Parking" zones (e.g., driveways, fire lanes), it instantly pushes a plate-specific Dashboard Anomaly Alert to the web app. This real-time notification empowers attendants to proactively address the exact violation. Concurrently, to protect these guaranteed reservation slots during peak congestion, the module automatically triggers an Overflow Protection Protocol

when the true availability of monitored sections drops to zero. This protocol routes all subsequent walk-in vehicles directly to the unmonitored Section F rough road area via redirection prompts printed on their dispensed QR tokens, securely logging all state transitions into the central database.

5.5.3 Dynamic Billing & Vendor Validation Module

This module governs the time-based financial logic and third-party validations of the parking lifecycle, processing all fee computations while the vehicle's physical location status securely remains 'PARKED'. The billing sequence initiates when a customer scans their original QR token at the exit scanner, prompting the backend database to calculate the exact elapsed duration. Based on this computation, the system dynamically routes the transaction through one of several strict billing scenarios: stays under 15 minutes trigger a free grace period (updating Payment_Status to 'BYPASSED'); standard daytime parking incurs a flat ₱30.00 fee ('PAID'); and vehicles remaining past 23:00 or rolling over into a new calendar day automatically trigger a ₱100.00 overnight fee ('PAID_OVERNIGHT'). For pre-planned stays, the system enforces a strict digital reservation pricing matrix: a base fee of ₱60.00 per day, mathematically stacking an additional ₱100.00 overnight surcharge plus the subsequent ₱60.00 daily fee for any bookings spanning two days or crossing the overnight threshold. To support the facility's commercial ecosystem efficiently and eliminate redundant paper waste, the module handles vendor validations by allowing retail tenants to simply scan the customer's original entry QR code. This digital handshake directly updates the token's database status to 'PAID_VENDOR', granting the customer a strict 20-minute exit window. Finally, the system securely handles exceptions via a Lost Ticket protocol, where untracked customers are penalized with a ₱100.00 surcharge on top of their accrued fees. Ultimately, every state change generates a transaction receipt and a tamper-proof financial audit record, completely eliminating manual calculation errors and revenue leakage.

5.5.4 Exit Verification & State Finalization Module

This module acts as the facility's secure physical gatekeeper, enforcing the "Closed-Loop" architecture to prevent ticket-swapping fraud and unauthorized exits. The finalization sequence is triggered

when a customer approaches the exit and gets their license plate captured by the exit-facing CCTV CAMERA. In a case where the LPR fails, a customer will then scan their Quick Response (QR) token at the exit terminal as a fallback. Upon receiving these inputs, the Python backend queries the centralized SQLite database using the scanned QR ID to retrieve the vehicle's current Payment_Status and the original visual string (Logged_Plate) captured during entry. The system then executes a strict decision matrix: it verifies that the financial obligation is cleared (status equals 'PAID', 'BYPASSED', 'PAID_VENDOR', 'PAID_OVERNIGHT' or 'PAID_RESERVATION') and performs a comparative logic check to ensure the newly extracted Camera_Plate exactly matches the Logged_Plate. Crucially, to accommodate the physical limitations of motorcycles, the algorithm gracefully bypasses the visual check if the Logged_Plate is explicitly recorded as NULL, relying solely on the validated QR status. If the dual-token verification succeeds, the system updates the database Location_Status to 'EXITED', automatically signals the microcontroller relay to actuate the exit boom barrier, and asynchronously updates the spatial allocation module to reflect the freed parking slot. Conversely, if the transaction is 'UNPAID' or a plate mismatch occurs, the hardware relay maintains a closed state, effectively mitigating revenue leakage.

5.5.5 Centralized Database & Auditing Module

This module operates as the system's "Single Source of Truth," functioning as a highly secure, relational SQLite database that underpins all transactional and spatial logic within the facility. By continuously aggregating data streams from the entry and exit cameras alongside physical QR token scans and vendor validations, the module systematically enforces strict billing rules, such as grace periods and overnight thresholds, by calculating the exact duration of each parking session. Crucially, it inextricably links the finalized Payment_Status (e.g., standard, vendor-validated, or overnight) with the Dual-Token identifiers—the generated QR ID and the visual license plate string—to create an immutable audit record. This centralized architecture directly resolves the facility's legacy vulnerabilities by completely eliminating manual "notebook" recording errors, closing the "Open Loop" physical security flaw, and rectifying the historical absence of entry auditing. Furthermore, by maintaining a persistent, real-time ledger of both

physical vehicle occupancies and financial clearances, the module equips the facility administrator with irrefutable proof of daily revenue and precise, data-driven capacity trends, effectively transforming raw operational inputs into actionable business intelligence.

Building upon this centralized data architecture, the module extends its functionality into a comprehensive analytics and reporting layer that transforms raw transactional data into structured financial and operational insights. For the financial side, the system generates a Revenue Summary that consolidates total earnings across daily, weekly, and monthly periods, providing administrators with clear visibility into overall performance. Revenue streams are further segmented into reserved slot income, overnight parking charges, and regular parking fees, allowing for granular analysis of income sources. The module also generates reports that identify peak revenue hours, enabling operators to understand high demand periods, while calculating the average revenue per vehicle to assess pricing efficiency and utilization profitability. Complementing this, the system supports trend and comparative analysis, offering evaluations of daily, weekly, and monthly performance metrics. It distinguishes peak versus off-peak earnings, highlighting demand fluctuations, and computes growth rates to indicate increases or declines in revenue over time. These insights empower management to make data-driven adjustments to pricing strategies, staffing, and capacity planning.

In terms of reservations, the module provides a detailed Reservation Summary, tracking the total number of bookings across different timeframes. It categorizes reservations into active, completed, and no-show instances, enabling administrators to evaluate user reliability, optimize slot allocation, and minimize revenue losses caused by unused reservations. From an enforcement perspective, the system maintains a robust Violations Summary, recording all infractions on a daily, weekly, and monthly basis. Violations are classified into specific types such as illegal parking, non-payment, and unauthorized slot usage, while also identifying repeat offenders through historical pattern tracking. This structured categorization enhances both operational enforcement and policy evaluation. Finally, the module includes a Penalty Summary that

aggregates all financial penalties issued within the system. It tracks total fines imposed, collected payments, and outstanding balances, providing a clear view of enforcement effectiveness and revenue recovery. Additionally, the system computes the average fine amount, offering insights into penalty consistency and deterrence strength.

Together, these reporting capabilities elevate the module beyond a transactional database into a fully integrated decision-support system, enabling administrators to monitor financial health, enforce compliance, and continuously optimize parking operations with precision and accountability.

5.5.6 Web Application with Data Analytics and Cashier Module

This module operates as the system's secure, role-based frontend, bridging the autonomous hardware logic with human-in-the-loop oversight through three specialized user interfaces. Page 1 (Attendant Cashier Interface) empowers operational staff to manually process financial clearances and actively enforce facility rules. Attendants can query vehicle records via QR scans or direct license plate inputs to process standard cash or e-payments; crucially, this interface securely integrates the Lost Ticket protocol, automatically calculating accrued duration fees alongside the requisite penalty surcharge for un-ticketed vehicles. Upon clicking "Confirm Payment," the system rigidly logs the specific Staff_ID for accountability and updates the database state to 'PAID' to satisfy the exit barrier logic, while simultaneously functioning as the active monitoring hub that receives real-time Spatial Anomaly Alerts whenever the vision module detects unauthorized parking.

Page 2 (Administrative Analytics Dashboard) serves as the executive business intelligence hub for the facility owner, aggregating all transactional data to visualize revenue by payment type, tracking manual cashier logs to permanently eliminate "Open Loop" revenue leakage, and generating vision-backed Capacity Heatmaps to optimize staff deployment during peak congestion.

Page 3 (Tenant Vendor Portal) provides commercial partners with a streamlined, isolated interface; rather than dispensing physical drive-thru coupons, vendors simply utilize this page to scan a customer's original entry token, instantly updating the database state to 'PAID_VENDOR' and securely initiating the 20-minute exit window while completely eliminating redundant paper waste.

Finally, Page 4 (Customer Page) allows the customer to create an account to register their license plate into the system so that the LPR cameras' will recognize the plate number if they have a reservation. The web app will also be the primary way for the customer to reserve a slot. The customer will be able to select which slot they want to reserve, from Zone B-E, and for how long (1-3 days). Zone A will be reserved for accounts that are confirmed to be PWD citizens and any VIP personnel entered into the system by the administrator. Next, the web app pulls live data to display the real-time availability of parking slots to the user. Before even arriving at the facility, users can open the app to see exactly how many slots are open in Zones A, B, C, D, or E. Once a user enters the facility and the Vision System assigns them a slot, the app dynamically updates to guide the user to their designated space. Lastly, customers can check the web app if they have any outstanding bills for violations committed inside the parking lot. The customer can enter their license plate into the web app and it will show any violations that the customer has not paid yet.

5.6 Data Specification

5.6.1 Entity Relationship Diagram

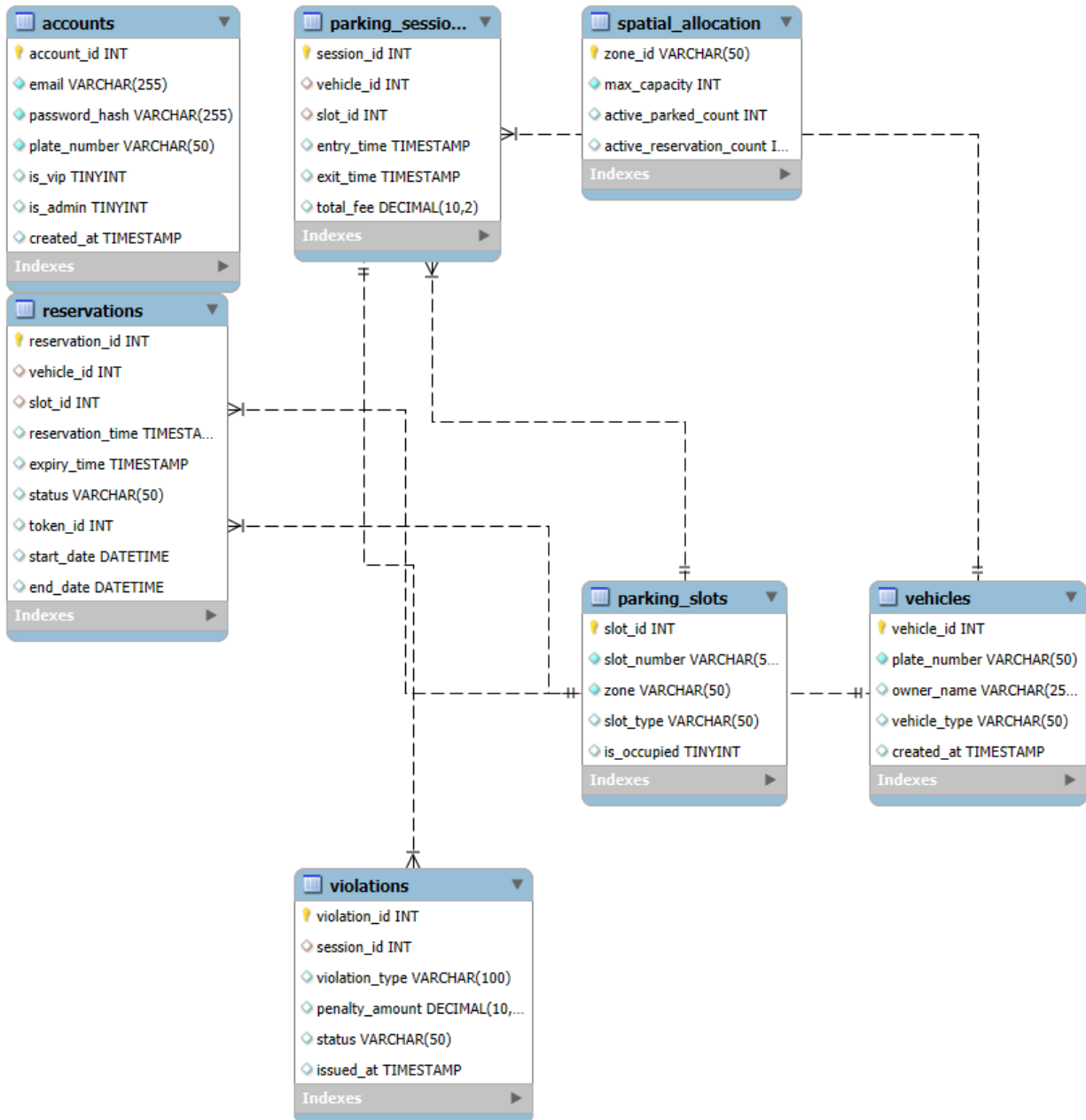


Figure 5.1 Entity Relationship Diagram

The Entity Relationship Diagram illustrates the database structure of the smart parking system and the relationships between its core operational entities. Designed as a "Single Source of Truth," the architecture enforces system resilience by decoupling physical license plate identification from token-based digital access.

The Vehicles entity serves as a central operational anchor, holding unique plate numbers, owner details, and vehicle types. Linked directly to vehicles on a one-to-one basis is the Accounts entity, which manages user credentials and privilege levels, such as VIP or PWD status, enabling rule-based access to restricted areas like Zone A regardless of camera detection performance.

The Reservations entity tracks prepaid parking windows and holds the QR token lifecycle completely independent of camera hardware. When a vehicle enters or exits, the Parking Sessions entity records real-time physical stays by linking the corresponding vehicle to a specific space in the Parking Slots entity, capturing entry and exit timestamps and calculating total fees.

For policy enforcement, the Violations entity attaches directly to active parking sessions to log infractions, penalty amounts, and payment statuses whenever overstays or unauthorized zone usage occur. Finally, the Spatial Allocation entity acts as a high-level summary module, tracking zone-level capacity and balancing real-time parked counts against active reservations to prevent overbooking and optimize facility flow.

5.6.2 Tables

5.6.2.1 accounts

Column Name	Datatype	Description
account_id	INT	unique identifier for the user account.
email	VARCHAR(255)	Unique email address used for user login and communication.
password_hash	VARCHAR(255)	Securely hashed user password for authentication.
plate_number	VARCHAR(50)	Plate number of the vehicle
is_vip	TINYINT	VIP or PWD privilege flag
is_admin	TINYINT	Admin flag
created_at	TIMESTAMP	Record creation timestamp

5.6.2.2 vehicles

Column Name	Datatype	Description
vehicle_id	INT	unique identifier for the vehicle.
plate_number	VARCHAR(50)	Plate number of the vehicle

owner_name	VARCHAR(255)	Full name of the vehicle owner or driver
vehicle_type	VARCHAR(50)	Classification of the vehicle
created_at	TIMESTAMP	Record creation timestamp

5.6.2.3 parking_slots

Column Name	Datatype	Description
slot_id	INT	unique identifier for the parking slot
slot_number	VARCHAR(50)	Unique code or label identifying the slot
zone	VARCHAR(50)	Parking zone or section designation
slot_type	VARCHAR(50)	Category of the slot
is_occupied	TINYINT	Physical occupancy status

5.6.2.4 reservations

Column Name	Datatype	Description
reservation_id	INT	unique identifier for the reservation
vehicle_id	INT	Foreign key referencing the associated vehicle in the vehicles table
slot_id	INT	Foreign key referencing the reserved slot in the parking_slots table
reservation_time	TIMESTAMP	Timestamp when the reservation request was created
expiry_time	TIMESTAMP	Timestamp when the active reservation window expires
status	VARCHAR(50)	Lifecycle state of the booking/token

token_id	INT	Unique identifier for the digital QR code token generated for fall-back entry.
start_date	DATETIME	Scheduled starting date and time of the parking reservation
end_date	DATETIME	Scheduled ending date and time of the parking reservation.

5.6.2.5 parking_sessions

Column Name	Datatype	Description
session_id	INT	unique identifier for the parking stay
vehicle_id	INT	Foreign key referencing the parked vehicle in the vehicles table
slot_id	INT	Foreign key referencing the occupied space in the parking_slots table
entry_time	TIMESTAMP	Timestamp recorded when the vehicle enters through the barrier gate.
exit_time	TIMESTAMP	Timestamp recorded when the vehicle exits through the barrier gate
total_fee	DECIMAL(10,2)	Calculated fee for the duration of the parking stay.

5.6.2.6 violations

Column Name	Datatype	Description
violation_id	INT	unique identifier for the violation record.
session_id	INT	Foreign key referencing the specific session in parking_sessions that incurred

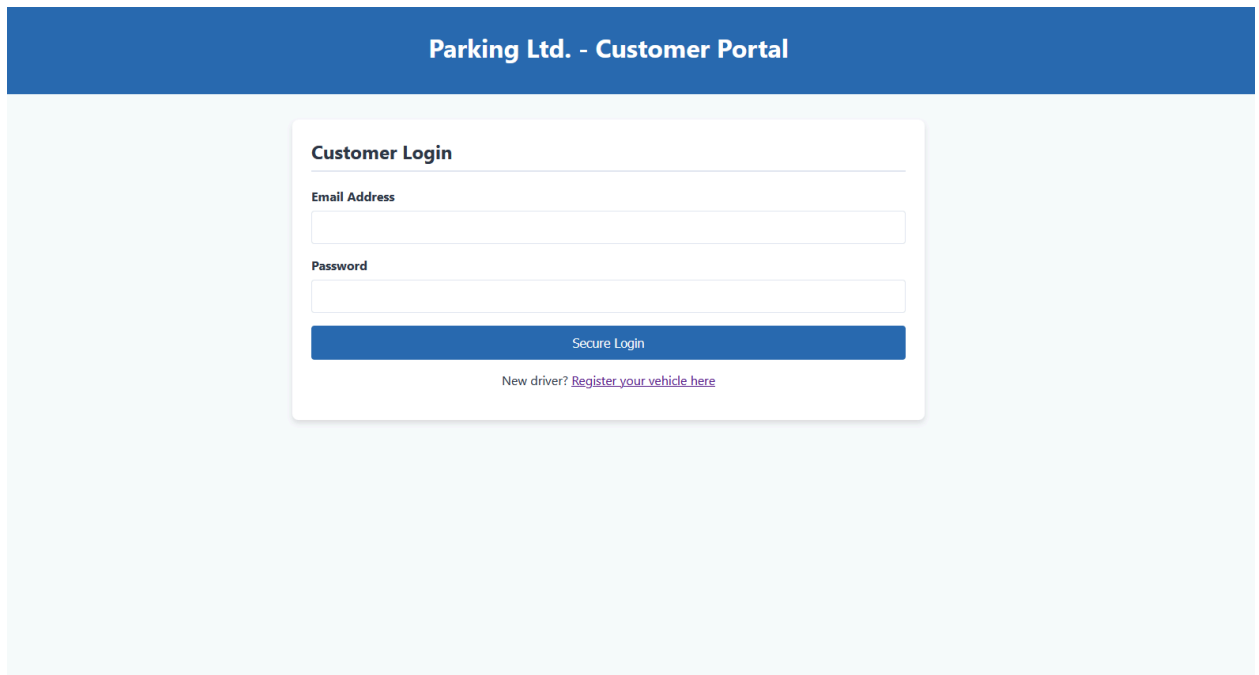
		the penalty
violation_type	VARCHAR(100)	Description or category of the penalty
penalty_amount	DECIMAL(10,2)	fine charged for the committed penalty
status	VARCHAR(50)	Payment/settlement status of the violation fine
issued_at	TIMESTAMP	Timestamp when the violation was detected and recorded

5.6.2.7 spatial_allocation

Column Name	Datatype	Description
zone_id	VARCHAR(50)	unique code representing a physical parking area or zone.
max_capacity	INT	Total number of available physical slots configured for the zone.
active_parked_count	INT	Real-time count of vehicles currently occupying slots in the zone
active_reservation_count	INT	Real-time count of valid, non-expired reservations assigned to the zone.

5.6.3 Screen Specifications

5.6.3.1 Login



The screenshot displays the 'Parking Ltd. - Customer Portal' login interface. It features a blue header bar with the text 'Parking Ltd. - Customer Portal'. Below the header, a white login form is centered on a light blue background. The form is titled 'Customer Login' and contains two input fields: 'Email Address' and 'Password'. A blue button labeled 'Secure Login' is positioned below the password field. At the bottom of the form, there is a link that reads 'New driver? [Register your vehicle here](#)'.

This page allows customers to log in to the Parking Ltd. portal using their registered email address and password. The interface contains input fields for login credentials, a submission button to authenticate the user, and a link directing new users to the vehicle registration page.

5.6.3.2 Register

Parking Ltd. - Customer Portal

Create an Account

Email Address

Password

Vehicle License Plate (One per account)

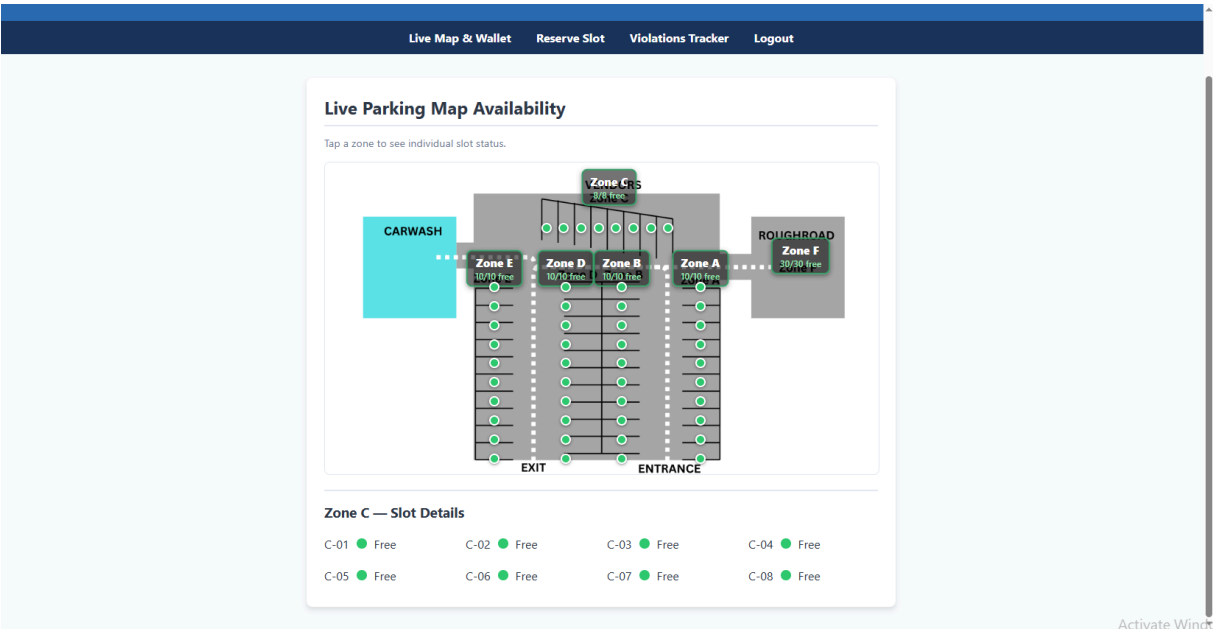
e.g. ABC-1234

Register Vehicle & Account

Already have an account? [Login here](#)

This page enables new drivers to create an account by providing an email address, setting a password, and linking a unique vehicle license plate number. The interface includes account creation fields, a registration button, and a navigation link to return to the login page

5.6.3.3 Dashboard



This page serves as the primary hub for logged-in customers to monitor real-time parking availability across zones and access their digital entry pass. The interface features a live grid showing free and occupied parking slots.

5.6.3.4 Slot Reservation

Parking Ltd. - Customer Portal

Live Map & Wallet

Reserve Slot

Violations Tracker

Logout

Reserve a Parking Slot

Select Zone

Zone A (Standard)

Start Date

14 / 08 / 2026

End Date

14 / 08 / 2026

Total Fee: ₱60.00

Confirm Payment & Book

Parking Ltd. - Customer Portal

Live Map & Wallet

Reserve Slot

Violations Tracker

Logout

Reserve a Parking Slot

Your Digital Entry Token



Slot Allocation: Zone B - B-02

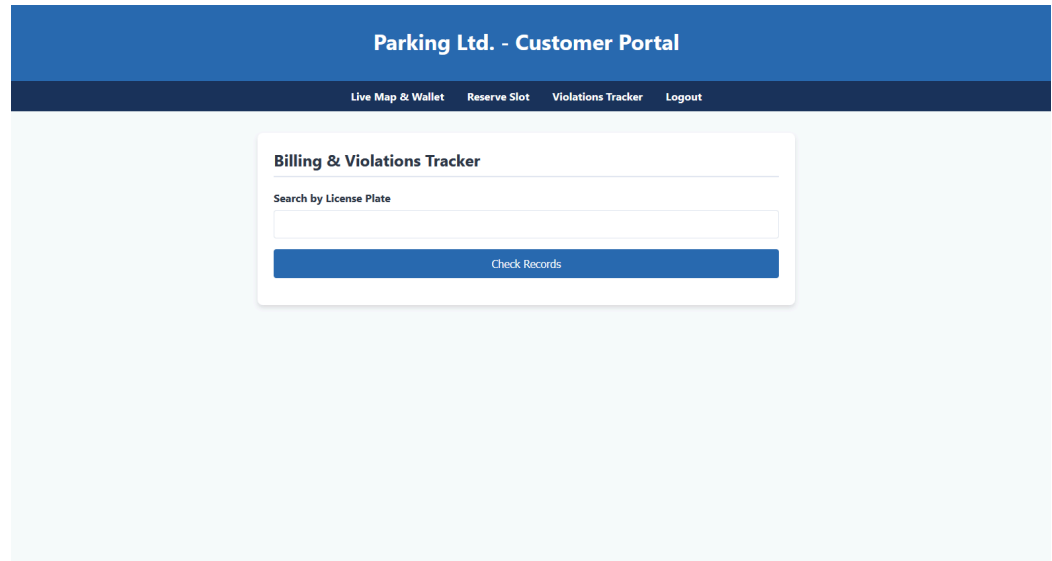
Reservation Dates: 2026-07-29 to 2026-07-30

Total Fee: ₱220.00

This page allows customers to book a parking slot in advance by selecting a preferred zone (including standard or PWD/VIP options) and setting start and end dates. The interface dynamically

calculates the reservation fee, provides a confirmation button, and displays a digital entry QR code token once a reservation is successfully booked

5.6.3.5 Violations Tracker



The screenshot displays the 'Parking Ltd. - Customer Portal' interface. At the top, a dark blue header contains the title 'Parking Ltd. - Customer Portal' and a navigation bar with links: 'Live Map & Wallet', 'Reserve Slot', 'Violations Tracker', and 'Logout'. The main content area is light blue and features a white card titled 'Billing & Violations Tracker'. Inside this card, there is a section labeled 'Search by License Plate' with a text input field and a blue button labeled 'Check Records'.

This page allows customers to search and view outstanding parking violations associated with their vehicle. The interface includes a search input field for entering a license plate number and a dedicated results container that displays penalty amounts, violation types, issuance timestamps, and payment statuses

5.6.3.6 Admin Portal

Parking Ltd. - Admin Portal

System Overview

Live Map & Wallet

Reserve Slot

Violations Tracker

Logout

System Status & Overview

Total Capacity

78 Slots

Occupied

0 Slots

Available

78 Slots

Active Violations

0 Cases

Ongoing Reservations

1 Active

Ongoing Reservations

All currently active reservations. Overriding cancels the reservation and frees the assigned slot.

Plate	Account	Zone / Slot	Start	End	Fee	Actions
NFK3620	testing@email.com	Zone C - C-01	2026-08-14	2026-08-14	₱60.00	Override / Cancel

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Account Management & VIP Status

Search by email or plate number and toggle VIP status (VIP accounts can reserve Zone E).

Email	Plate	VIP Status	Role	Actions
blank@gmail.com	BLK1234	Standard	User	Make VIP
demo@gmail.com	XYZ1234	Standard	User	Make VIP
gab@gmail.com	GAB1234	Standard	User	Make VIP
new@email.com	NEW 1234	Standard	User	Make VIP
rorocruz@gmail.com	YXZ1234	Standard	Admin	Make VIP
test@gmail.com	ABC-1234	Standard	User	Make VIP
testing@email.com	NFK3620	Standard	User	Make VIP
testvip@email.com	NFB5514	VIP	User	Remove VIP

Live ALPR Feed & Detection Logs

Real-time vehicle detection via YOLOv8 & ALPR.

Timestamp	Plate Number	Assigned Slot	Confidence	Status
No active detections found.				

Slot Management & Manual Overrides

Zone A

10 / 10 available

A-01 — Available Force Occupied	A-02 — Available Force Occupied	A-03 — Available Force Occupied	A-04 — Available Force Occupied	A-05 — Available Force Occupied
A-06 — Available Force Occupied	A-07 — Available Force Occupied	A-08 — Available Force Occupied	A-09 — Available Force Occupied	A-10 — Available Force Occupied



This page serves as the administrative command center for system oversight and manual override controls. The interface displays high-level analytics cards (total capacity, occupied slots, available slots, and active violations), a real-time ALPR detection log table, and an interactive slot management grid allowing administrators to manually override slot occupancy statuses.

5.7 Proposed System Advantages

Automating entrance and exit operations with ALPR (Automated License Plate Recognition) and QR digital tokens slashes recurring labor costs while eliminating entry bottlenecks, human math errors, and cash leakage.

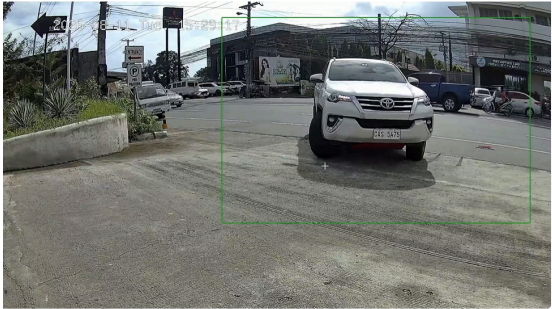
Key Functional Advantages Over Human Staff

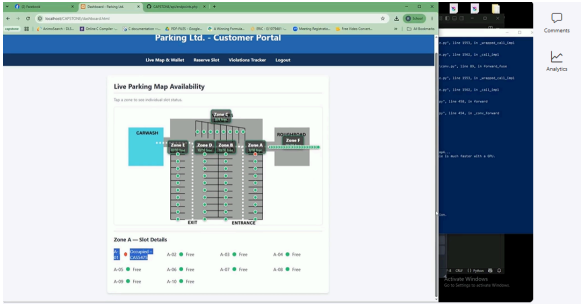
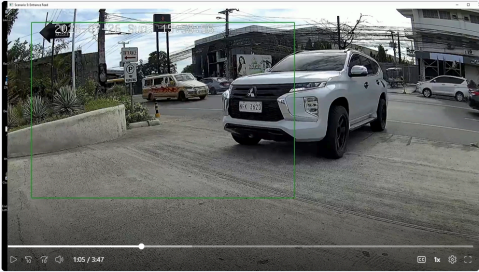
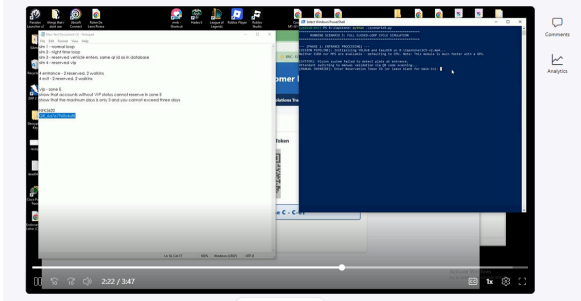
- Zero Cash Handling & Automated Leakage Prevention:** Human attendants managing physical cash often encounter disputes over entry times or miscalculated fees. The system automatically computes exact stay

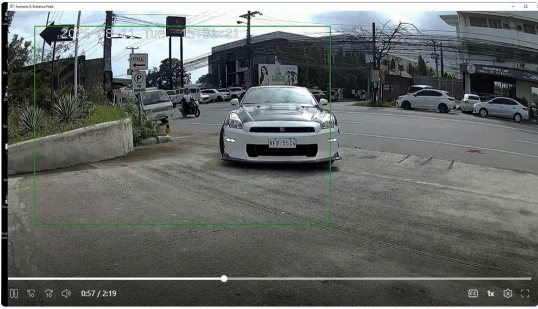
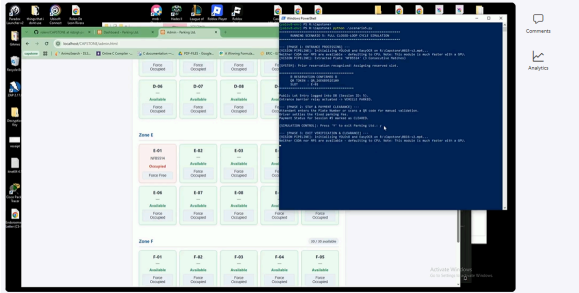
duration down to the minute, applies overnight surcharges, handles lost ticket penalties, and tracks payment status without manual intervention.

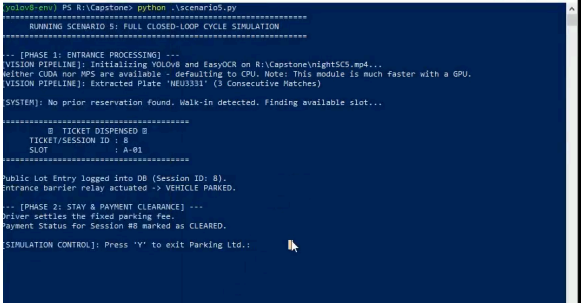
- **Pre-Reserved Slot Guarantee:** Personnel at a gate can only tell drivers if the lot is full or empty as a whole. The system lets users pre-reserve specific slots and guarantees a spot before the driver even leaves home.
- **Automated Violation & Enforcement Tracker:** Rather than relying on attendants walking rows to manually log expired tickets or unauthorized parking, the backend cross-references active parking sessions against slot sensors and automatically flags violations for targeted enforcement.
- **Elimination of Queue Bottlenecks:** Physical tickets slow down traffic during peak entry and exit hours. ALPR camera scanning matches license plates instantly against active database sessions, lifting barrier gates with zero driver friction.
- **Data-Driven Administrative Control:** Manual attendants provide basic headcounts at best. The admin system delivers immediate visibility into exact occupancy percentages, real-time ALPR detection logs, peak usage hours, and forced slot overrides for maintenance.

5.8 Scenarios

Scenario / Mode	Operational Context	System Behavior & Logic Flow
1. Standard	Regular walk-up driver without a prior booking.	ALPR Entry: Camera captures plate number on entry loop trigger. Database Logic: Checks plate against active database records. If clear, creates a temporary session with a dynamic token and opens the gate. Billing: Normal hourly rates apply upon exit. 

		
2. Reserved	Pre-booked driver with an existing slot allocation.	ALPR Entry: Camera scans plate and matches it to an active and valid reservation in the database. Database Logic: Bypasses temporary entry creation; links session directly to assigned zone and slot. The screen displays the assigned spot. Billing: Applied pre-paid rate structure (no double charging at exit).  

3. VIP	Pre-registered high-priority or subscriber account	ALPR Entry: Camera identifies registered plate. Database Logic: System bypasses standard pricing tables, auto-allocates designated VIP zone slots, and logs access without charging standard gate fees. User Experience: Instant gate lift with custom screen welcome message.  
4. Night	Low-ambient-light conditions (6 PM – 6 AM).	Hardware Adjustments: Infrared (IR) illuminators auto-engage on cameras to mitigate vehicle headlight glare. Database Logic: Automatically triggers overnight surcharge calculations via time checks.

		 
5. Rain	Heavy precipitation, water distortion, and lens droplets.	CV Processing: Image processing pipeline switches to high-contrast filtering and multi-frame averaging to read wet/dirty plates.

Reference/s:

Al-batat, R., Angelopoulou, A., Premkumar, S., Hemanth, J., & Kapetanios, E. (2022). An end-to-end automated license plate recognition system using YOLO based vehicle and license plate detection with vehicle classification. *Sensors*, 22(23), 9477. <https://doi.org/10.3390/s22239477>

Rani, R., Kumar, S., Pippal, S. K., Gund, M., Chaudhari, U., Agrawal, R., Dalsaniya, M., & Verma, L. (2024). IPS: Intelligent Parking System using YOLO and image processing. *International Journal of Transportation Development and Integration*, 8(3), 447-453. <https://doi.org/10.18280/ijtidi.080308>

Prasetyawan, P., Aulia, M. A. C., Utami, N. S., & Ramadhani, U. A. (2023). Automatic License Plate Recognition (ALPRON) using optical character recognition method. *Journal of Applied Informatics and Computing*, 9(3). <https://doi.org/10.30871/jaic.v9i3.9903>

Zakaria, M. R., Safei, S., Wan Yussof, W. N. J., & Fitri, S. (2025). Enhancing parking systems with QR code-integrated automatic license plate recognition through convolutional neural networks. *Journal of Advanced Research Design*, 131(1), 117-125. <https://doi.org/10.37934/ard.131.1.117125>

Park, S.-H., Yu, S.-B., Kim, J.-A., & Yoon, H. (2022). An all-in-one vehicle type and license plate recognition system using YOLOv4. *Sensors*, 22(3), 921. <https://doi.org/10.3390/s22030921>

Satya, B., Manongga, D., Hendry, & Aminuddin, A. (2025). Optimized YOLOv8 for automatic license plate recognition on resource constrained devices. *Engineering, Technology & Applied Science Research*, 15(2), 21976-21981. <https://doi.org/10.48084/etasr.9983>

Connie, L., Kim On, C., & Patricia, A. (2018). A review of automatic license plate recognition system in mobile-based platform. *Journal of Telecommunication, Electronic and Computer Engineering (JTEC)*,

10(3-2), 77-82.

Sassa, A. C., de Almeida, I. A., Pereira, T. N. F., & de Oliveira, M. S. (2023). Scrum: A systematic literature review. *International Journal of Advanced Computer Science and Applications*, 14(4).

Verwijs, C., & Russo, D. (2022). A theory of scrum team effectiveness. *ACM Transactions on Software Engineering and Methodology*, 32(3), 1–51. <https://doi.org/10.1145/3571849>

Alami, A., & Krancher, O. (2022). How Scrum adds value to achieving software quality? *Empirical Software Engineering*, 27(7), 165. <https://doi.org/10.1007/s10664-022-10208-4>

Ezginci, Y., & Karadaş, M. (2019). Access Control and Recording System Using SQLite and RFID. *International Journal of Scientific and Technological Research*.

Pawlaszczyk, D. (2022). SQLite. In *Mobile Forensics–The File Format Handbook: Common File Formats and File Systems Used in Mobile Devices* (pp. 129-155). Cham: Springer International Publishing.

Lin, T., Rivano, H., & Le Mouél, F. (2017). A survey of smart parking solutions. *IEEE transactions on intelligent transportation systems*, 18(12), 3229-3253.

Al-Turjman, F., & Malekloo, A. (2019). Smart parking in IoT-enabled cities: A survey. *Sustainable Cities and Society*, 49, 101608.

MarkLines. (2025, January 15). Philippines - New car sales in 2024 by vehicle type. MarkLines Automotive Industry Portal.

https://www.marklines.com/en/statistics/flash_sales/automotive-sales-in-philippines-by-month-2024

Technavio. (2026, January 2). Smart parking market size to grow by USD 15.19 billion from 2024 to 2029.

Technavio. <https://www.technavio.com/report/smart-parking-market-industry-analysis>

Kirovska, N., & Koceski, S. (2015). Usage of Kanban methodology at software development teams. *Journal of applied economics and business*, 3(3), 25-34.

Patil, S., & Patil, A. (2019). QR code based smart parking system. 2019 International Conference on Vision Towards Emerging Trends in Communication and Networking (ViTECoN), 1-5.
<https://ieeexplore.ieee.org/document/9182180>

Atlassian. (2026, February 17). Kanban | Atlassian. <https://www.atlassian.com/agile/kanban>

Yadav, R., Mittal, M. L., & Jain, R. (2019). Lean practices in software development projects: A literature review. *AIP Conference Proceedings*. <https://doi.org/10.1063/1.5123966>

Atlassian. (2026, February 17). What is Lean Methodology? | Atlassian.
<https://www.atlassian.com/agile/project-management/lean-methodology>

Al-Turjman, F., & Malekloo, A. (2019). Smart parking in IoT-enabled cities: A survey. *Sustainable Cities and Society*, 49, 101608. <https://doi.org/10.1016/j.scs.2019.101608>

Du, S., Ibrahim, M., Shehata, M., & Badawy, W. (2012). Automatic license plate recognition (ALPR): A state-of-the-art review. *IEEE Transactions on circuits and systems for video technology*, 23(2), 311-325.

Lin, T., Rivano, H., & Le Mouél, F. (2017). A survey of smart parking solutions. *IEEE transactions on intelligent transportation systems*, 18(12), 3229-3253.

Redmon, J., Divvala, S., Girshick, R., & Farhadi, A. (2016). You only look once: Unified, real-time object detection. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 779-788).

Luque-Vega, L. F., Michel-Torres, D. A., Lopez-Neri, E., Carlos-Mancilla, M. A., & González-Jiménez, L. E. (2020). lot smart parking system based on the visual-aided smart vehicle presence sensor: SPIN-V. *Sensors*, 20(5), 1476.

Tagalanao, D. A., Caguete, J. M., & Macatulad, E. (2024). Development of an automatic data-centered parking space occupancy detection with vehicle license plate web-based monitoring. *Journal of Computational Innovations and Engineering Applications*, 6(1), 28–34.
<https://www.dlsu.edu.ph/wp-content/uploads/pdf/research/journals/jciea/vol-6-1/6tagalanao.pdf>

Land Transportation Office. (2018). Standardization of motor vehicle license plates. Department of Transportation, Republic of the Philippines. <https://lto.gov.ph>

Republic Act No. 4136. (1964). Land Transportation and Traffic Code. Official Gazette of the Republic of the Philippines. <https://www.officialgazette.gov.ph/1964/06/20/republic-act-no-4136/>

Toll Regulatory Board. (2023). Vehicle classification. Department of Transportation, Republic of the Philippines. <https://trb.gov.ph/faqs/vehicle-classification.html>

Parmar, J., Das, P., & Dave, S. M. (2020). Study on demand and characteristics of parking system in urban areas: A review. *Journal of Traffic and Transportation Engineering (English Edition)*, 7(1), 111-124.
<https://doi.org/10.1016/j.jtte.2019.09.003>